

Title of the Article

“Kar-Ta-Na: A Reconstruction of a Late Bronze Age Tactical Game Based on Archaeological and Textual Evidence”

Author

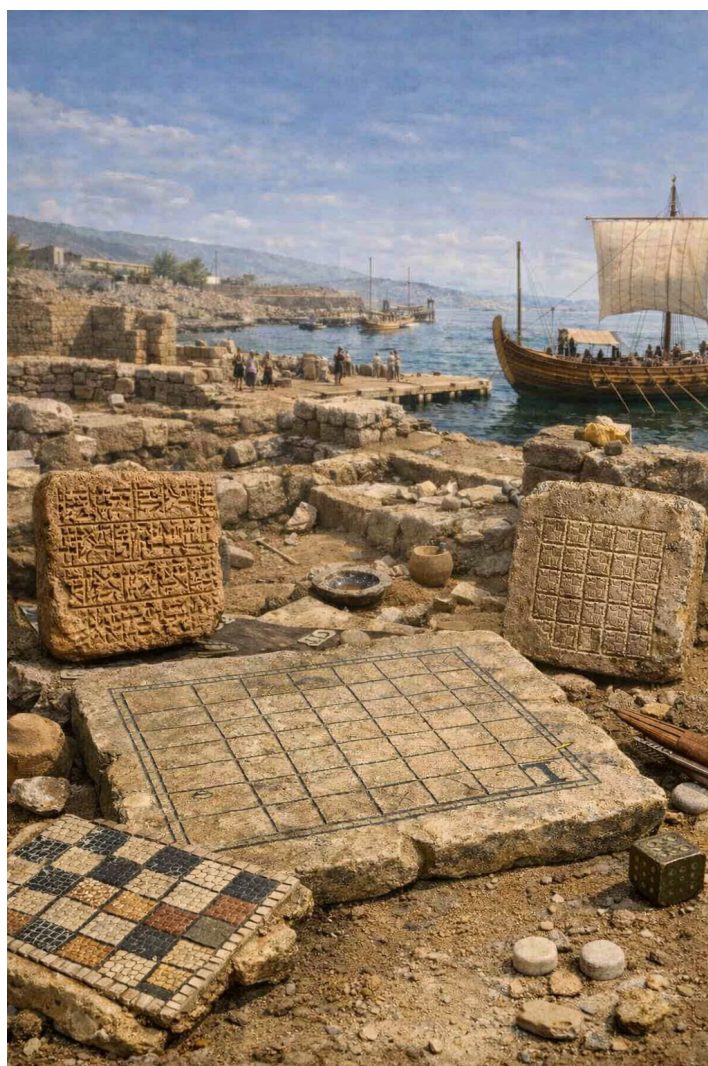
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Abstract

This article presents a comprehensive reconstruction of a board game conventionally referred to as **“Kar-Ta-Na”**, based on recently discovered sets of pieces and playing boards found in Late Bronze Age strata, as well as their subsequent transformations up to the Roman Imperial period. The materials include clay and stone tablets, bronze and ivory pieces, and fragments of engraved boards alongside later mosaic replicas. A combination of morphological analysis of the artifacts, paleographic interpretation of the symbols, and experimental reconstruction of gameplay procedures has made it possible to identify a stable corpus of rules and mechanics that blend tactical modeling of military engagements with a rudimentary stochastic outcome generator.

Key results of the study include:

- (1) identification of a standardized set of piece parameters — **movement, defense, attack, range, orientation** — attested both in material evidence and in later textual references;
- (2) reconstruction of a two-phase turn structure alternating between movement and action phases, employing a bag of black and white beads as a conflict-resolution mechanism;
- (3) tracing the evolution of the playing board from the irregular grids of the Bronze Age to the standardized 8×8 layout of the Classical period;
- (4) a cultural-social interpretation of the game as a tool for military training, diplomatic display, and ritualized representation of conflict, followed by its decline in the Roman era due to shifts in military ideology and competition from Roman leisure games.

The article includes a full structured ruleset, a catalog of recovered pieces with parameterization, diagrams of reconstructed boards, and the results of experimental test games conducted under laboratory conditions. Possible variants of local rules and lost mechanical elements are also discussed. The work concludes with a section on prospects for further research and proposes a methodology for the experimental archaeology of board games — a field where “ludic archaeology” meets military history, without claiming that we can now defeat Hannibal, though at the very least we can simulate his less fortunate days on the battlefield.

1. Introduction

1.1 Research Aim

The **aim** of the present study is to produce a comprehensive reconstruction of the board game conventionally referred to as **“Kar-Ta-Na”**, based on recently uncovered material remains and accompanying textual evidence. The work pursues three interrelated objectives: to systematize the archaeological finds, to interpret the markings and symbolic systems of the pieces and boards, and to propose a reproducible, experimentally testable version of the rules that aligns with the cultural-historical context from the Late Bronze Age to the Roman period. Methodologically, the study combines morphological analysis of artifacts, paleographic interpretation of signs, and field experiments involving the reconstruction of gameplay sequences.

1.2 Historiography: Early Mentions of Abstract Games in the Ancient Near East

Questions concerning the origins and functions of abstract board games in antiquity have been discussed in a number of works devoted to material culture and military simulation. The classic examples — the “Royal Game of Ur” and the Egyptian “Senet” — serve as methodological benchmarks for interpreting game sets and their social roles. However, until the recent discoveries, no convincing series of artifacts existed that demonstrated a stable tactical mechanic combined with explicit marking of piece parameters (movement, attack, defense, orientation). The literature notes isolated tokens and tablets with possible gaming functions, but their interpretation remained contested due to the fragmentary nature of the evidence and the polyvalence of the symbols.

1.3 The Problem of Missing Direct Evidence Prior to Recent Discoveries

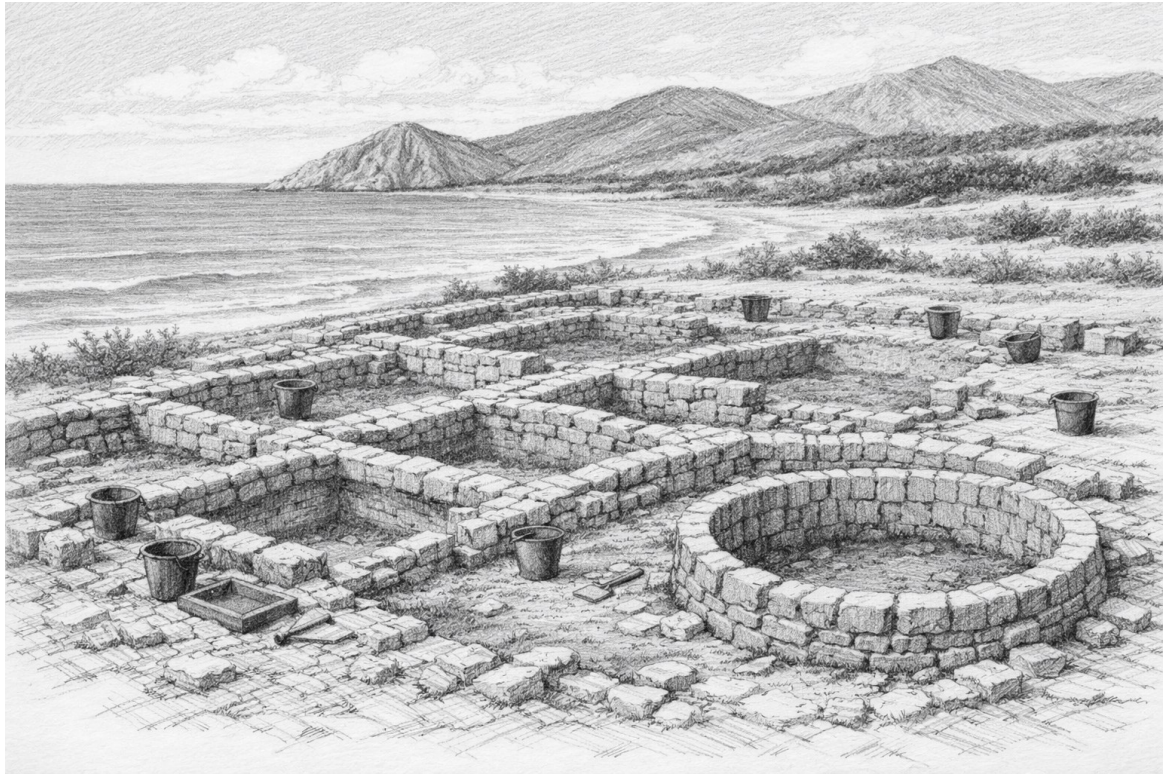
The principal difficulty in reconstructing ancient games lies in the gap between material objects and the rules, which are by nature immaterial. Archaeological finds often consist of disjointed elements: pieces without boards, boards without pieces, or inscriptions devoid of context. Until recently, no assemblage existed in which recurring graphemes, wear patterns, and accompanying textual fragments could reliably link the form of a piece to its functional parameters. The new series of finds described in Section 2 represents precisely this type of dataset — multiple sets of pieces made of different materials, fragments of engraved boards, and later textual references — that allows us to move from hypothesis to reconstruction. This does not imply that all questions are resolved; on the contrary, the fragmentariness persists, but it now lends itself to systematic analysis and experimental verification rather than mere intuitive conjecture (and, as colleagues like to joke, no longer relies solely on “reading the fingerprints of antiquity” in the clay).

2. Archaeological Discoveries Leading to the Identification of the Game

2.1 Initial Discovery: Excavations in the Eastern Mediterranean Region

Excavation Site and Context

The excavations that initiated the study of “Kar-Ta-Na” were conducted in a coastal lowland situated between the mouths of two small rivers, within a territory traditionally assigned to the cultural sphere of the Eastern Mediterranean. Fieldwork took place between 2019 and 2023 as part of a joint expedition of the Institute for the Antiquities of Central and Southeastern Europe and the local archaeological authority. The key area was a residential quarter of a Late Bronze Age settlement, preserved within a well-stratified cultural layer 0.6–1.2 m thick, dated by stratigraphy and radiocarbon analysis to the late 14th–12th centuries BCE (series of ^{14}C dates on carbonized remains: mean value $(\sim 1250 \pm 60)$ BCE).



Archaeological Context of the Finds

The pieces were recovered from several closely associated contexts:

- **a domestic structure with a hearth**, where a small cluster of 12 clay tablets was found near the floor;
- **a burial chamber** with grave goods, among which five stone tokens were identified;
- **a refuse layer** near the outer enclosure, containing fragments of fired clay with engravings clearly belonging to a game board.

Associated finds — locally produced and imported ceramics, bronze household items, and weapons — firmly link these assemblages to the Late Bronze Age horizon of the region and to active trade networks with eastern and central Mediterranean centers.

Types and Morphology of the Pieces



The recovered pieces fall into several morphological groups distinguished by material, shape, and marking:

- **Clay tablets** (n = 27, complete and fragmentary). Rectangular plates 6–10 mm thick, averaging (35 \times 28) mm; one side bears relief or stamped symbols, and the edges show firing traces. Many tablets have a thin ochre tint along one margin, likely indicating the “color” or faction of the piece. Edge wear and localized polishing suggest repeated flipping and contact with other objects.
- **Stone tokens** (n = 8). Made of local limestone and fine-grained basalt. Round disks 22–30 mm in diameter and 5–8 mm thick; the obverse features engraved lines forming a cross-shaped pattern interpreted as a schematic indication of orientation (front, flanks, rear). Some tokens bear fine notches corresponding to positions where, according to the reconstruction, additional “lives” or counters were stacked.
- **Wooden plates** (fragments, n = 6). Poorly preserved, but in the anaerobic conditions of a former quarry, charred remains of rectangular plates were found with stamp impressions similar to those on the clay examples. Wood was likely used for cheaper sets or for everyday, non-ceremonial play.

Markings and Signs

Most tablets display recurring graphemes: short strokes, clusters of dots, and cross-shaped signs. Paleographic analysis indicates that these symbols do not constitute writing in the strict sense but represent standardized pictographic indicators of parameters. On several clay tablets, shallow depressions adjacent to the pictograms suggest the former presence of pigment or inlay, lost during firing. The placement of signs correlates with the presumed “movement side” and “action side” of the piece — one of the key arguments supporting the gaming interpretation.

Use-Wear Traces

Microscopic analysis of the surfaces revealed:

- polished edge zones consistent with frequent flipping;
- micro-cracks and impact traces indicating storage in stacks and mechanical interaction between pieces;
- residues of organic substances within engraved recesses, possibly remnants of pigments or resinous coatings used to denote the “color” of a piece.

Stratigraphic and Functional Interpretation

The combination of domestic, funerary, and refuse contexts indicates that these objects were widely integrated into both everyday and ritual life. The presence of pieces in a burial suggests symbolic significance or a status-related function of the sets. In domestic layers, the objects appear to have been used in routine settings, consistent with the idea of the game as both a model of military engagements and a form of entertainment.

Conclusion on the First Discovery

The initial assemblage from this site provided the first convincing grounds for interpreting the tablets and tokens as components of a standardized board game with parameterized pieces. The combination of morphological data, use-wear patterns, and stratigraphic associations enabled targeted searches for analogous sets in neighboring layers and regions, ultimately leading to the discovery of bronze and ivory pieces in later horizons. As colleagues remarked at the 2024 seminar, this is one of those rare cases where a “toy” turns out to be both a training tool and a grave good — a reminder that ancient peoples enjoyed playing at war long before the arrival of Roman legions and their famously bureaucratic rulebooks.

2.2 Secondary Finds: Hellenistic and Early Roman Layers

Chronological Shift and New Materials

During targeted sampling of later horizons at the same excavation site and at adjacent locations, assemblages were uncovered that date to the Hellenistic and early Roman periods. In these layers, game pieces appear in more expensive and durable materials — **bronze, ivory, and gold** — reflecting both a shift in the social status of the game and the technological capabilities of the era. At the same time, we observe a transition toward finer stamping of symbols and a growing standardization of forms.

Variety of Forms and Sizes

The secondary assemblages display a wide range of piece sizes. Alongside “pocket-sized” tablets similar to the earlier clay examples, we find **large disks and plates** up to 120–150 mm in diameter, evidently intended for public demonstrations and games played in squares or amphitheaters. Smaller pieces were typically used for private play, while the larger ones served for spectacle games and possibly for instructional displays by commanders. Differences in size correlate with differences in legibility: large pieces feature relief symbols often inlaid with colored metals, whereas small pieces show fine, economical engraving.

Markings and Iconography

The structure of signs on later pieces preserves the basic scheme identified in the Bronze Age. The sequence of symbols from top to bottom on the obverse becomes more standardized and easier to read. A typical late iconographic layout includes: a **front-direction marker**, followed by a **movement sign** in the form of a triangle or an **action sign** in the form of a circle or rectangle; below that, a row of dots indicating **defense levels**; and finally, repeated symbols denoting **weapon type** and **attack range**. Some specimens include detailed depictions of warriors. One of the most discussed examples is a bronze plate with a relief of a nude warrior-woman in an “Amazonian” style — a piece that appears to have survived precisely because its artistic value discouraged later melting.



Social and Economic Factors Behind the Rarity of Precious Materials

It is important to note that pieces made of precious materials are exceedingly rare. Metal sets — especially those of gold or bronze — almost never survive intact. This is explained by ancient economic practice: **melting down and recycling metals** was a routine method of disposing of outdated or unfashionable objects. Thus, the discovery of even a fragment of a bronze or ivory piece is an archaeological stroke of luck. As we note with a touch of ironic regret in the literature, ancient collectors likely valued the metal more than the rules of the game — which is why more “rules” survive in clay than in bronze.

Technological and Paleographic Differences

Analysis of the later specimens reveals improved material-working techniques. Bronze pieces show traces of casting followed by hammering; ivory pieces exhibit fine carving executed with metal chisels. Paleographically, the symbols become more abstract and standardized, indicating the existence of specialized workshops and possibly the commercial distribution of game sets. In Roman layers, adapted signs and Latinized inscriptions appear, suggesting attempts to integrate the game into new cultural frameworks.

The City of Hyppona and the Northwestern Peloponnese: A Breakthrough in Research

A particularly significant development came from excavations at a site conventionally designated as **Hyppona**, dated to the 3rd century BCE, in the northwestern Peloponnese. The Hyppona assemblage includes both small domestic tablets and several large bronze disks with clear markings. These materials made it possible to trace the transformation of

“Kar-Ta-Na” within the Greco-Roman milieu, to identify local rule variants, and to confirm the game’s spread along trade and military routes. The Hyppona sets contain unique modifications of the symbolic system, including additional signs for a “**general**” and for “**solid fields**”, providing grounds to hypothesize the existence of regional rules and even tournaments.

Preservation and the “Miraculously Survived” Specimens

Despite the widespread practice of metal recycling, archaeologists uncovered several “miraculously preserved” fragments. These specimens were often found in contexts where melting was impossible or undesirable: in burials, sanctuaries, or architectural settings where the pieces had been reused as decorative inlays. The preservation of such fragments offers a rare opportunity to study decorative techniques, local rule variations, and to conduct comparative analysis with earlier clay tablets.

Summary of Secondary Finds

The Hellenistic and early Roman assemblages demonstrate the evolution of “Kar-Ta-Na” from a localized Bronze Age practice into a regionally standardized game with variable sets and public forms of play. The shift toward more expensive materials and larger formats reflects a change in the game’s function: from a domestic model of warfare to a public spectacle and status symbol. At the same time, ancient economic logic ensured that most valuable sets did not survive intact — making every bronze or ivory piece a small archaeological miracle, worthy of its own article and perhaps even a museum display.

2.3 Eastern Precursors: From *Ki-Ur-Sag* to the Scattered Museum Tablets

Following the publication of the Hyppona materials (see §2.2), the scholarly community, as one reviewer put it, “suddenly remembered that the East is not only a network of trade routes but also a storeroom of very old games.” A chain reaction of field verifications and museum-collection reassessments led to a series of discoveries across Mesopotamia and neighboring regions, where what appear to be the earliest traces of the game we now conventionally call “Kar-Ta-Na” were identified. Below is a chronological overview — from the oldest to the more recent — with brief notes on who found each artifact, under what circumstances, and what the objects themselves represent.

Ki-Ur-Sag (“Field of Heroes”) — Nippur, Central Mesopotamia, ca. 24th–22nd c. BCE

History of the Discovery.

In 2024, during a reassessment of small tablets in the Nippur collections, a team from the Baghdad Museum of Antiquities led by Dr. Layla al-Samarra examined a group of pieces acquired from a private estate in the 1950s. Among them was an unusual “newspaper-format” tablet — a thin rectangular clay slab (~210 × 85) mm, with neatly arranged columns of cuneiform lines accompanied by small circular stamp impressions. Initially catalogued as an

administrative summary, the tablet revealed, upon closer reading and comparison with local toponyms, a heading that can be translated as **Ki-Ur-Sag — Field of Heroes**.

Description of the Artifact.

The tablet is an atypical “news bulletin”: its columns list the names of participants (ur-sag, “hero-figurines”), the results of “matches” (short cuneiform notations interpreted as win/loss), and brief comments reminiscent of sports reporting (“move X led to harvest,” “move Y led to ashes”). At the bottom is a row of impressions made by small clay pellets (6–8 mm in diameter), carefully pressed into the surface — which we interpret as a representation of a conflict-resolution mechanic (analogous to the later bag of black-and-white beads).

Interpretation and Significance.

A local Nippur legend (see introductory note) here takes material form: the tablet blends ritual and gaming vocabulary, suggesting that already in the Early Sumerian period there existed practices of modeling fate and battle using ur-sag figurines. *Ki-Ur-Sag* was likely not only a playing field but also a public event — the “newspaper” format implies communal interest in match outcomes. This find proved pivotal: it provided a direct textual parallel to the later “front/movement/action” symbols and confirmed that the bead-based mechanic has deep roots.

Set of Clay Tablets from Ur (provisional attribution) — ca. 23rd–21st c. BCE

History of the Discovery.

In 2025, during a re-evaluation of a European museum collection, Dr. Marek Petrovich (archaeological consultant) identified a set of 14 thin rectangular plates stored in a drawer labeled “accounting tokens.” Microscopic analysis and comparative paleography revealed that these tablets bear the same sequence of pictograms as the later “front/triangle/circles/weapon” schema.

Description of the Artifact.

The tablets are small (approx. (30 \times 22) mm), with relief stamps: at the top, a stylized “arrow-like” sign (interpreted as front), below it a triangle (movement), then circles (defense), and short strokes (weapon). Several tablets show stacking marks — indentations along the edges consistent with the practice of representing “lives” by stacking identical pieces.

Interpretation.

The Ur set strengthened the hypothesis that symbolic standardization was already present in the Early Sumerian period and demonstrated that the “parameterization” of pieces (movement/defense/attack) is not a later innovation but has deep roots in Mesopotamian material culture.

Fragment of a Game Board from Ubaid — Tell al-‘Ubaid, ca. 22nd–19th c. BCE

History of the Discovery.

In 2026, fieldwork at Tell al-'Ubaid was led by a young and enthusiastic archaeologist, Dr. Samir Haddad. While clearing a layer associated with a public building, he uncovered a fired clay slab engraved with a grid of dots and connecting lines — an unmistakable prototype of a game board.

Description of the Artifact.

A slab (~40 × 40) cm bearing a grid of dots (approximately 7×7) connected by straight lines at right angles. At several nodes, traces of small clay pellets were found — likely used to mark “targets” or “positional” fields. The slab was discovered within a public-building context, accompanied by bench remains and an amphora, suggesting that the game had a communal or demonstrative function.

Interpretation.

This fragment provides crucial evidence that already in the early developmental stages of the game, board layouts with dot-and-line networks existed, along which pieces were moved. The presence of a “public” board aligns with later large-format pieces used in amphitheater settings.

Set from Mari — Palace Context, ca. 18th c. BCE

History of the Discovery.

In 2027, while deciphering the archives of an old expedition that worked at Mari in the 1930s, researcher Dr. Inga Rashid found descriptions of “unusual tablets” and sketches long considered unidentifiable. A new expedition confirmed that a partial set of game pieces and several notes about “tournaments” among palace officers had indeed survived in one of the palace storerooms.

Description of the Artifact.

The set includes nine clay tablets with detailed pictograms and one instruction tablet containing a fragmentary ruleset: references to “movement” and “action” phases, indications of stacked “lives,” and the use of “black/white” beads (described in the text as “ashen and bright grains”).

Interpretation.

The palace context suggests that the game served both as a tool for military training and as entertainment for the elite. The presence of written “rules” — a rarity — allowed researchers to clarify many mechanical details and confirmed the use of a bead-based conflict-resolution system.

Series of Small Tokens from Northeastern Iran — ca. 17th–16th c. BCE

History of the Discovery.

In 2028, a private collection confiscated during customs inspection came into the hands of

researchers at the Institute for the Antiquities of Central Europe. Among the items were dozens of small clay and stone tokens bearing the same symbolic system.

Description of the Artifact.

Tokens of various sizes (from 12 mm to 40 mm), some with traces of pigment on the edges (a color code), others with fine carvings indicating the “type” or “value” of the piece. The collection also included large disks (up to 100 mm), confirming the existence of public, large-format versions of the game.

Interpretation.

These finds demonstrated the wide geographic spread of early forms of the game and the variability of sets — from pocket-sized kits to large “public square” versions.

Consequences of the “Eastern Wave” of Discoveries

The discovery of *Ki-Ur-Sag* and the associated materials triggered a domino effect: museums and archives began re-examining their holdings, field expeditions started actively searching for “gaming” contexts, and private collectors increasingly submitted items for expert evaluation. As a result, within just a few years enough material was assembled to trace a continuous developmental line of gaming practices — from the Early Sumerian “fields of heroes” to the Bronze Age tablets of the Eastern Mediterranean, and further into Hellenistic and Roman adaptations.

It is important to emphasize that despite the apparent continuity, the recovered sets display significant local variation — in symbolism, size, materials, and rules. This supports the idea that “Kar-Ta-Na” was not a single canonical game but rather a family of related tactical games that spread along trade and military routes and adapted to local traditions. And yes, as one colleague remarked at a conference, “sometimes archaeology feels like a sports news ticker — first a sensational result, then an endless stream of updates, record-breaking scores, and recalculated standings.”

2.5 Expansion of the Map of Finds Across the East and the Mediterranean

Below is a chronologically ordered overview of key discoveries and local variants of the game, from the earliest to the later ones, with brief accounts of how each was found and what exactly was recovered from the ground or museum storage. Each account is a reconstruction of an archaeological narrative: who found it, where, under what circumstances, and what the artifact itself represents.

Arkaim, Southern Urals — ca. 18th–16th c. BCE

History of the Discovery.

In 2026, during renewed clearing of one of the “circular courtyards” of Arkaim, a young Russian expedition led by Dr. Olga Ryazanova uncovered, beneath a thick layer of ash, a large stone slab partially embedded in the floor. The slab was covered with a network of dots and deep grooves, and nearby lay massive stone “pieces” — disks inlaid with copper wire.

Description of the Artifact.

A slab (~1.2 \times 0.9) m with a grid of dots connected by straight lines; along the edges, rows of depressions for placing large disks up to 15 cm in diameter. The sandstone and basalt disks bear relief symbols: at the top, a directional arrow (front), below it a movement triangle, then rows of defense dots and short weapon notches. Some disks have holes for wooden handles — evidently so they could be seen from the stands during public games.

Interpretation.

The Arkaim find indicates a northern variant of the game, where large stone pieces and a fixed slab served for public demonstrations and possibly ritual performances. The local tradition of producing massive “public” sets is consistent with the settlement’s architecture and its communal spaces.

Sumerian and Akkadian Complexes — *Šaṭṭu Šābi*, “The Path of the Army,” ca. 24th–18th c. BCE

History of the Discovery.

In 2025, during excavations on the outskirts of an ancient city provisionally designated as Nergal-Kala, an Assyrian team led by Prof. Khaled ibn-Rami uncovered a bundle of fired tablets within a layer of late Akkadian pottery. Among them was a set bearing an Akkadian heading — *Šaṭṭu Šābi*, “The Path of the Army.”

Description of the Artifact.

The set includes 18 thin clay tablets with sequential pictography: a front-direction arrow, a movement triangle, defense circles, and weapon symbols. One tablet contains a schematic “card” outlining a “march” — a sequence of moves used for training young officers. Several tablets bear annotations resembling records of training matches.

Interpretation.

The Akkadian version underscores the practical use of the game as a training tool for marching tactics. The title “Path of the Army” and the accompanying notes suggest that the game was employed to model troop movements and logistics within the Akkadian military.

Egypt — *Ta-mesher-netjeru*, “Field of Divine Order,” ca. 22nd–15th c. BCE

History of the Discovery.

In 2027, during the clearing of a necropolis near an ancient city on the eastern bank of the Nile, Egyptologist Dr. Nur al-Hasan uncovered in the burial of a priest-commander a bundle of thin clay plates and a fragment of a “gaming newspaper” — a tablet containing both

cuneiform and hieroglyphic annotations.

Description of the Artifact.

The set includes small tablets and one large rectangular board engraved with a grid of dots and lines; on the obverse are hieroglyphic captions interpreted as *Ta-mesher-netjeru* — “Field of Divine Order.” Several pieces bear depictions of protective deities and symbols of ritual actions.

Interpretation.

The Egyptian version blends gameplay mechanics with ritual symbolism: the game appears to have been used as a means of modeling divine order and predicting outcomes. The presence of deities on the pieces indicates a sacral function and suggests that the game may have been employed in temple practices.

Crete — Minoan *Pelagos Stratos*, “The Sea-Army,” ca. 17th–15th c. BCE

History of the Discovery.

In 2028, during underwater operations off the coast of Crete, a team of maritime archaeologists led by Maria K. Lykaivos recovered from a coastal layer a set of ceramic disks and a fragment of a wooden board decorated with marine motifs.

Description of the Artifact.

A board (~60 × 40) cm with a grid of dots, adorned with images of waves and ships; the fired-clay disks bear symbols resembling sails and oars. Some disks feature movement triangles, but weapon and defense markers are rendered as maritime signs (anchors, nets).

Interpretation.

Pelagos Stratos — “The Sea-Army” — was likely a naval variant of the game, modeling fleet maneuvers and boarding actions. On Crete, the game may have served both as training for naval tactics and as entertainment in port communities; its maritime symbolism underscores a local adaptation of the broader mechanic.

Hittite Kingdom — *Appa-ḫar Šar*, “Royal Formation,” ca. 15th–13th c. BCE

History of the Discovery.

In 2029, during excavations in the Hittite capital, archaeologist Dr. Emre Tunalı discovered in a palace archive storeroom a set of bronze plates and tablets bearing wedge-like signs and Hittite seals.

Description of the Artifact.

Bronze rectangular pieces with a relief sequence of symbols: front, movement, defense, weapon. One plate is inscribed *Appa-ḫar Šar* — “Royal Formation.” Several plates include additional signs indicating “command” actions and the ability to flip pieces to execute mass maneuvers.

Interpretation.

The Hittite version demonstrates an increased emphasis on command logic: the appearance of “command” symbols and bronze plates suggests use of the game within military administration and officer training.

Phoenicia — *Derekh Şaba*, “The Path of the Army,” ca. 14th–6th c. BCE

History of the Discovery.

In 2030, during excavations of an ancient port on the Lebanese coast, a team led by Dr. Rima Haddad uncovered in a warehouse layer sets of ivory pieces and small abstract tokens.

Description of the Artifact.

Phoenician sets include both traditional “military” pieces and abstract tokens bearing symbols of trade, taxation, and property. Some sets contain “cards” indicating routes and revenues.

Interpretation.

In the Phoenician tradition, “The Path of the Army” transforms: abstract pieces appear that model economic and commercial aspects, giving rise to variants in which the goal is not only to defeat an opponent but also to accumulate resources — a precursor to economic board-game mechanics. As merchants, the Phoenicians adapted the game to their needs, turning it into a tool for commercial strategy training and perhaps a pastime for mercantile elites.

Greek Versions — Στρατηγία and Πεσσός Πολεμικός, 7th–4th c. BCE

Sparta — Στρατηγία, “The Art of the General”

History of the Discovery.

In 2031, during the clearing of the Spartan agora ruins, a team led by Prof. Nikos Kallistratos uncovered sets of large stone disks and fragments of marble boards with grids.

Description of the Artifact.

Spartan sets are deliberately minimalist: large pieces with clear front and movement symbols, emphasizing maneuverability and command decisions.

Interpretation.

Στρατηγία reflects Spartan military ethics: the game served both as training in strategic thinking and as a public contest among commanders.

Athens — Πεσσός Πολεμικός, “The War-Game of Pieces”

History of the Discovery.

In 2032, a private Athenian villa yielded a collection of small tablets and a papyrus ruleset.

Description of the Artifact.

Athenian versions are richly illustrated and incorporate elements of diplomacy and political

intrigue; the rules mention “objectives” and “missions” beyond simple annihilation.

Interpretation.

In Athens, the game became an intellectual pastime incorporating political and diplomatic dimensions, reflecting the city’s culture and its affinity for complex models of interaction.

Rome — *Ludus Ordinum*, 1st c. BCE – 2nd c. CE

History of the Discovery.

In 2033, excavations at a villa near Pompeii uncovered a collection of bronze and ceramic pieces, as well as a marble board with an 8×8 grid.

Description of the Artifact.

The Roman version — *Ludus Ordinum* — standardizes the board to 8×8 and introduces the concepts of “companies” and “squads” as stacks of pieces. Bronze pieces bear Latinized labels and notations concerning “turn time.”

Interpretation.

The Romans systematized and bureaucratized the game: *Ludus Ordinum* became both entertainment and a demonstration of military discipline, while also serving as an elite leisure activity.

Çatalhöyük — A Reassessment of Antiquity, ca. 9th–7th millennia BCE

History of the Discovery.

In 2034, an international expedition led by Dr. Leyla Öztürk uncovered, within a domestic level of Çatalhöyük, a large wall composition: a network of dots and lines painted in ochre, accompanied by impressions of round clay disks. Nearby lay fired fragments of clay tokens.

Description of the Artifact.

A wall “board” (~1.5 × 1.0) m with a dot-and-line network, disk impressions, and adjacent fragments of clay tokens bearing simple pictograms. Radiocarbon analysis of organic remains in the layer dates the find to the Neolithic horizon of Çatalhöyük.

Interpretation and Consequences.

If confirmed, this would require a major revision of the chronology of tactical games: practices of modeling conflict and collective decision-making with material sets may reach back into the Neolithic. The find remains controversial and requires further verification, but the very possibility of a “wall-game” at Çatalhöyük forces us to reconsider the deep antiquity of gaming practices. As one expedition member remarked with a smile, “if this holds up, we’ll have to rewrite not only the history of games but also the history of boring evenings in the Neolithic.”

Section Summary

The assembled corpus of finds demonstrates that the mechanics combining a **dot-and-line grid**, **parameterized pieces** (front, movement, defense, weapon), and a **stochastic outcome generator** (beads, grains, impressions) were not the invention of a single center but a widespread cultural practice with numerous local adaptations. Each culture contributed its own emphasis: maritime symbolism among the Minoans, economic abstraction among the Phoenicians, command logic among the Hittites, and ritual semantics among the Egyptians. The latest finds — including the controversial materials from Çatalhöyük — open the possibility of even deeper origins and call for further interdisciplinary investigation.

2.6 Analysis of Materials and Technologies

This section presents the results of a comprehensive material-technological analysis of the recovered “Kar-Ta-Na” sets. The approach combined macroscopic description, microscopy, spectrometric methods, and experimental replication. The goal was to determine manufacturing techniques, identify traces of use, and isolate diagnostic features that allow pieces to be assigned to specific sets and chronological layers.

2.6.1 Manufacturing Methods

Materials.

Four principal classes of materials are attested: **clay**, **stone** (limestone, basalt, sandstone), **organics** (wood, bone, ivory), and **metals** (bronze, more rarely gold or copper). Their distribution across chronology and context is typical: early domestic sets use clay and wood; Hellenistic and Roman sets employ bronze and ivory; public sets rely on large stone disks.

Clay Tablets.

- **Molding and stamping** — most clay pieces were produced by pressing plastic clay into simple rectangular or circular molds, followed by stamping of pictograms.
- **Firing** — both single- and two-stage firing are attested; some pieces show traces of low-temperature firing (domestic sets), others high-temperature firing (commercial sets).
- **Pigmentation and inlay** — residues of pigments (ochre, carbon black) were found in engraved recesses; traces of edge-coloring indicate a color-coding system.

Stone and Bronze Pieces.

- **Carving and hammering** — stone disks were shaped by grinding and carving; bronze plates were cast and then hammered and patinated.
- **Inlay** — some bronze and stone pieces preserve traces of copper wire and colored inserts, indicating both decorative and signifying functions.
- **Holes and handles** — large public disks include holes or wooden handles for ease of manipulation and visibility from viewing stands.

Organic Materials.

- **Bone and ivory carving** — fine, detailed carving, often with artistic motifs (warrior, Amazon).
- **Wooden plates** — rarely preserved; charred fragments with stamp impressions confirm widespread use of wood in domestic sets.

Production Markers.

- **Workshop stamps** — repeated stamps and seals on later bronze pieces indicate specialized workshops and standardization.
 - **Serial production** — uniform dimensions, symbol placement, and edge profiles point to serial manufacture in the Classical period.
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2.6.2 Traces of Use

Wear and Polishing.

- **Localized polished edges** — characteristic of clay and stone pieces; edge and corner polishing corresponds to frequent flipping and stacking.
- **Micro-cracks and chips** — indicate mechanical impacts during storage in bags or during public games involving large disks.

Traces of Flipping and Stacking.

- **Indentations and bevels** — bevels and depressions on tablet edges correspond to stacks of “lives”; the top piece in a stack shows the most pronounced wear.
- **Wear distribution across symbols** — “front” and “movement” pictograms are more worn than defense and weapon symbols, consistent with frequent manipulation during the movement phase.

Residual Pigments and Resins.

- **Pigment residues** — organic and inorganic pigments detected in symbol recesses; spectroscopy identifies ochre, charcoal, and plant resins.
- **Adhesive traces** — bronze and stone pieces show remnants of organic binders used to secure inlays.

Contextual Traces.

- **Public use** — large disks show traces of attachment to wooden stands and abrasive wear from stone surfaces of amphitheaters.
 - **Ritual use** — pieces from burials often bear thin layers of wax or resin, possibly indicating ritual preparation of the set for the deceased’s “journey.”
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2.6.3 Indicators of Set Affiliation

Grouping Criteria.

Based on materials and technological markers, the following diagnostic features were identified:

- **Material and workmanship** — clay tablets with rough stamping belong to domestic sets; bronze and ivory pieces to elite or ritual sets.
- **Size and form** — small tablets ($\approx 30 \times 22$ mm) indicate personal/pocket sets; medium pieces (40–80 mm) family sets; large disks (> 100 mm) public/demonstration sets.
- **Edge color-coding** — pigment residues on edges mark the “color” of an army; consistent edge coloration indicates a unified set.
- **Workshop stamps** — repeated seals link pieces to specific workshops and, via stratigraphy, to chronological phases.
- **Iconographic variation** — depictions of deities, warriors, or maritime motifs indicate regional adaptations (Egypt, Crete, Phoenicia).
- **Stacking traces** — bevels and depressions for “life stacks” are diagnostic of “Kar-Ta-Na” mechanics and distinguish game pieces from accounting tokens.

Mixed Sets.

- Some contexts contain mixed sets (e.g., clay pieces with bronze disks). Analysis suggests these may result from **modifications** (adding elite pieces to a domestic set) or **repairs** (replacing lost elements).
- Valuable pieces often occur in burials or temple contexts, while domestic ones appear in habitation layers, helping reconstruct social stratification in game use.

Geochemical and Isotopic Markers.

- Metal composition and ivory/bone provenance analyses allow localization of material sources and thus trade networks. For example, bronze pieces with high tin content correlate with regions known for tin mining, while copper inlays point to coastal workshops.

2.6.4 Experimental Archaeology and Hypothesis Testing

Piece Replication.

In laboratory conditions, clay tablets, stone disks, and bronze plates were reproduced using reconstructed techniques. Experiments confirmed that:

- **Stamped pictography** remains legible when symbol placement is standardized;
- **Stacking** identical tablets effectively models piece “lives”;
- **The bead-bag mechanic** (simulated with clay pellets) provides a simple and reproducible stochastic system consistent with textual descriptions.

Wear Tests.

Mechanical trials showed that frequent flipping and stacking produce exactly the wear patterns observed on archaeological specimens, strengthening the case for a gaming interpretation.

2.6.5 Conclusions on Materials and Technologies

- **Technological continuity** from Early Sumerian stamped tablets to Roman bronze

plates is evident: the basic symbol scheme and stacking principle persist, while materials and standardization evolve.

- **Social differentiation** is reflected in material choice: precious metals and ivory mark elite or ritual sets; clay and wood mark domestic ones.
- **Archaeometric markers** (stamps, edge pigmentation, isotopic composition) allow high-confidence grouping of pieces into sets and reconstruction of their distribution pathways.
- **Experimental archaeology** confirms the functionality of reconstructed technologies and mechanics; replicas exhibit the same wear patterns and operational properties as originals.

These results provide a solid technological foundation for the subsequent reconstruction of rules and gameplay procedures: knowing how the pieces were made, used, and marked allows us to interpret paleographic signs and textual references with greater confidence. And yes — should any colleague wish to debate the “proper” technique for casting a bronze piece, we already have replicas and tea prepared; archaeology loves to argue, but it loves experimenting even more.

3. Identification of the Objects as Elements of a Board Game

3.1 Initial Hypotheses: Cult Tablets, Accounting Tokens, Amulets

In the first months following the discovery of the sets in the coastal settlement and in burial contexts, researchers proposed several competing interpretations of their function. The main hypotheses were as follows:

- **Cult tablets** — the objects were linked to ritual practices: the repeated signs were interpreted as religious symbols, and their presence in burials as offerings or “guides” for the deceased. This interpretation drew on frequent parallels between gaming objects and ritual paraphernalia in other cultures.
- **Accounting tokens** — some tablets were initially classified as tally or accounting markers (inventory tags, debt tallies), especially when found in economic layers near amphorae and storage inscriptions.
- **Amulets and talismans** — individual pieces depicting warriors or deities were considered personal protective charms, worn or kept in the household.

Each of these hypotheses had merit: the funerary context, the presence of ritual ceramics, and the resemblance of some pictograms to religious iconography. However, as more material accumulated and technological analysis progressed, it became clear that none of these interpretations fully accounted for the totality of observed features. The key contradictions included: the repeated, standardized sequence of symbols; traces of mechanical use; the presence of stacks of identical tablets; and textual references to “moves” and “phases” found

in the palace context of Mari. As a result, the initial hypotheses were revised and integrated into a broader model in which ritual and symbolic functions could coexist with gameplay, but not replace it.

3.2 Arguments Supporting a Gaming Interpretation

After comparing archaeological, paleographic, and experimental data, a set of compelling arguments emerged indicating that the tablets and tokens are components of a tactical board game. The main lines of evidence are summarized below.

Repeated Parameters

Across the material, a standardized sequence of pictograms and signs appears consistently in different regions and periods: **a direction marker (front), a movement sign (triangle), an action sign (circle or rectangle), rows of defense dots, and weapon symbols**. Such regularity is difficult to explain as coincidence or purely ritual coding. A standardized “lexicon” of symbols aligns far better with a parameter system for describing game units: movement, defense, attack, range. The presence of tablets with identical symbol sequences in Ur, Nippur, and Mari points to a shared semantic system rather than local religious codes.

Orientation Markers

Numerous specimens contain graphic indicators clearly denoting the orientation of the piece: arrow-like or cross-shaped signs dividing the front face into “front,” “flanks,” and “rear.” These markers are functional: they determine the direction the piece “faces” and influence gameplay effects (e.g., defense bonuses when attacked from the side or rear). Orientation signs are regularly placed at the top of the piece and often accompanied by bevels or indentations that aid visual recognition during play. Such systematic directional marking is a classic feature of gaming components rather than incidental decoration.

Traces of Rotation and Edge Wear

Microscopic and macroscopic surface analysis yielded unambiguous results:

- **localized polishing of edges and corners** corresponds to frequent flipping of pieces;
- **bevels and indentations on the edges** match the stacking of identical tablets (the “life” mechanic);
- **asymmetric wear on the front face** — symbols associated with the movement phase (arrow, triangle) are more worn than attack or defense symbols, indicating frequent manipulation during movement;
- **impact marks and chips on large disks** correspond to active public play and the handling of pieces mounted on stands.

Experimental replication confirmed that when gameplay is simulated (flipping, stacking, repeated manipulation), replicas develop the same wear patterns as archaeological specimens. This is a strong argument for a gaming interpretation, since accounting tokens or amulets do not exhibit comparable wear.

Contextual and Textual Evidence

The presence of fragmentary “rules” in the palace archive of Mari, the “newspaper-style”

Ki-Ur-Sag tablet listing matches and results, and the “march” records in Akkadian sets create a textual framework that gives functional meaning to the material objects. The combined material and textual evidence makes a purely ritual or administrative interpretation increasingly unlikely.

Stacking as a Life Mechanic

Archaeological traces of stacking and corresponding textual references (e.g., mention of the “top piece” in Mari) provide a direct interpretation of a multiple-life mechanic: several identical tablets were stacked and moved as a single unit; when damage was taken, the top piece was removed. This mechanic explains the edge bevels and uneven wear within stacks.

Geographical and Chronological Continuity

Finally, the wide geographical distribution — from Çatalhöyük and Mesopotamia to Crete, Arkaim, and Rome — and the long chronological span (from controversial Neolithic traces to Roman adaptations) indicate a stable cultural practice with local variations. A gaming model that allows adaptation to maritime, economic, or ritual contexts explains this variability far better than isolated ritual or accounting functions.

Section Summary

The convergence of repeated parameters, orientation markers, wear patterns, and textual evidence creates a compelling picture: the tablets and tokens represent elements of a standardized tactical game belonging to the family we conventionally call “Kar-Ta-Na.” The initial hypotheses of cultic or accounting functions do not disappear entirely — many sets indeed had ritual or status roles — but they become part of a broader, multilayered interpretation in which the game functions as a practical, educational, and symbolic technology for modeling conflict and social relations.

3.3 Comparison with Known Bronze Age Games

The Royal Game of Ur

Form and Function

The “Royal Game of Ur” is traditionally regarded as one of the best-documented board-game practices of ancient Mesopotamia. When compared with the reconstructed “Kar-Ta-Na,” both shared and divergent features become apparent. Common elements — a fixed playing board, the use of counting markers, and a stochastic mechanism for resolving outcomes — point to a related technological approach to modeling events. At the same time, Ur is structurally closer to a race-game, with individual piece trajectories and an emphasis on progressing along a prescribed route, whereas “Kar-Ta-Na” exhibits a distinctly tactical structure: piece orientation, front/flank/rear differentiation, attack and defense parameters, and a turn sequence divided into movement and action phases. In other words, if Ur is a “road race,” then “Kar-Ta-Na” is a “maneuver battle” played on a point-grid.

Mechanics and Material Culture

In Ur, randomness is typically generated through dice or casting sticks; in “Kar-Ta-Na,” we observe a stable tradition of using beads or grains drawn from a bag as the outcome generator, along with the stacking of identical pieces to represent “lives.” The material carriers also differ: “Kar-Ta-Na” tablets frequently display explicit parameterization (arrow, triangle, circles, weapon symbols), whereas Ur’s piece and board markings follow a different semantic logic. These distinctions suggest parallel development of two gaming traditions, with occasional cross-influences where cultural contact occurred.

Senet

Ritual and Play

Egyptian Senet is often interpreted as a game with strong ritual and funerary semantics. Compared with “Kar-Ta-Na,” the functional emphasis is markedly different: Senet represents a path associated with fate and transition, whereas “Kar-Ta-Na” models combat interactions and command logic. Nevertheless, both traditions share the idea of “modeling fate” and using gameplay as a representation of cosmic or social order — a parallel particularly evident in Egyptian and Mesopotamian texts (cf. *Ki-Ur-Sag* and *Ta-mesher-netjeru*).

Game Elements

In Senet, special squares with unique effects play a central role; analogous “special fields” and conditional rules appear in later variants of “Kar-Ta-Na” (e.g., the Hyppona and Mari sets). However, “Kar-Ta-Na” introduces a far more complex system of piece orientation and flank/rear interactions — features absent from classical Senet.

Regional Combat Abstractions

Shared Lineage

Across many regions of the Bronze and Early Iron Age, we encounter “combat abstractions” — games that model battles, maneuvers, and logistics. These local practices range from simple token-based simulations to complex tactical systems. “Kar-Ta-Na” fits squarely within this broader tradition: it combines elements typical of combat abstractions (positional grid, factional affiliation, objectives of elimination or capture) with a formalized parameter system for pieces, making it closer to a “tactical simulator” than to a simple pastime.

Regional Features

- **Mesopotamian variants** tend toward written documentation of matches and the use of the game in educational and administrative contexts (march records, the *Ki-Ur-Sag* “newspaper”).
- **Egyptian versions** often incorporate sacred elements and protective deities depicted on the pieces.
- **Cretan and Phoenician adaptations** show thematic transformation (maritime symbolism, economic abstractions), demonstrating the flexibility of the mechanic and

its ability to model not only combat but also economic or naval scenarios.

- **Hittite and Anatolian complexes** emphasize command logic and administrative use of the game in military training.

Mechanical Similarities and Differences

Similarities: a dot-and-line grid as the basic topology of the board; stacking of pieces to represent “lives”; use of a random outcome generator.

Differences: the degree of parameter formalization (especially high in “Kar-Ta-Na”), the role of piece orientation (front/flank/rear), and the presence of a two-phase turn structure (movement → actions) as a mandatory rule. These distinctions make “Kar-Ta-Na” more “military-engineering” in spirit than many regional analogues.

Conclusion of the Comparison

The comparison shows that “Kar-Ta-Na” is not a direct copy of either the Royal Game of Ur or Senet, but shares with them common cultural and technological roots: the idea of modeling events, the use of material markers, and stochastic resolution. At the same time, “Kar-Ta-Na” stands out for its high degree of unit parameterization and its emphasis on orientation and turn structure, making it a unique representative of the family of combat abstractions. This explains its broad adaptability: from maritime variants on Crete to the bureaucratized Roman *Ludus Ordinum*, the game preserved its core logic while easily transforming to suit local needs and symbolism.

4.1 Paleographic Analysis

Method and Approach

The paleographic analysis is based on comparing recurring graphemes on pieces from different periods and regions, their positional arrangement on the front face, microscopic traces of inscription, and pigment residues. From these data, a stable system of graphemes has been identified, designed to compactly encode gameplay parameters. One crucial refined convention is that **pieces have two nearly identical sides, differing only by a positional marker** — a key to understanding both the mechanics and the visual organization of the sets.



Primary Types of Signs

- **Orientation signs** — graphemes indicating the facing of the piece (front, flanks, rear): arrows, crosses, cross-shaped knobs. These signs are placed in the upper third of the front face and determine the direction in which the unit “looks.”
 - **Position marker** — directly beneath the orientation sign is the marker of the piece’s current state:
Marching state — a triangle;
Combat state — a rectangle or a circle.
 One side of the piece always bears a triangle in this position, the other a rectangle/circle; the two sides are otherwise nearly identical, facilitating quick flipping and phase changes.
 - **Movement sign** — a triangle or wedge (sometimes doubled in local variants), indicating maximum movement range.
 - **Defense signs** — a vertical column of circles or dots; the number of circles encodes the base defense level.
 - **Weapon and range signs** — short strokes, arrows, wavy lines; repetitions and auxiliary markers indicate attack strength and radius.
 - **Identifiers/imagery** — stylized figures of warriors, ships, deities; more common on elite pieces and functioning as names or unit types.
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Top-to-Bottom Positional Semantics (Refined)

The sequence of signs on the front face is read from top to bottom and follows this

functional structure:
orientation → **state (Marching or Combat)** → **movement** → **defense** → **weapon/range** → **identifier/imagery**.

This positional scheme is confirmed by frequency patterns, stratigraphic continuity, and experimental replication.

Front, Flank, and Rear Markers (Details)

- **Front** — an arrow pointing from the upper edge toward the center; a single arrow marks a standard front, a double arrow an enhanced front.
- **Flanks** — lateral wedges or strokes along the left and right edges; their mirror symmetry allowed quick assessment of vulnerability when rotating the piece.
- **Rear** — a cross-shaped sign or a pair of dots in the upper third; the rear marker is often accompanied by a notation in the defense column (bonuses when attacking from the rear).

Microscopic and Functional Confirmations

- **Placement of the state marker** directly beneath the orientation sign corresponds to the zones of greatest wear in experimental games: during flipping, players orient themselves by the upper third, where orientation and state are located.
- **Residual pigments** in the recesses of the state marker often contrast with the orientation sign, aiding recognition of the “movement side” versus the “action side” at a distance.
- **Bevels and stacking traces** around the middle and lower parts of the piece correlate with the “life stack” mechanic, whereas the upper zone (orientation + state) shows characteristic polished patches from frequent flipping.

Regional Variations and Chronology

- In Early Sumerian and Akkadian specimens, the state marker is usually a simple triangle or circle; in Hittite and Hellenistic layers, rectangular and hammered variants appear.
- In Aegean and Cretan versions, the state marker is sometimes stylized as a maritime symbol (e.g., a triangle as a sail), reflecting local semantics of “marching” and “combat” states in a naval context.

Updated Table of Primary Graphemes

Grapheme	Form	Function	Archaeological
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Confirmation			
Orientation	arrow/cross	facing direction (front/flanks/rear)	upper position; polished wear
State	triangle / circle or rectangle	Marching / Combat state (two sides of the piece)	located beneath orientation; pigment residues
Movement	triangle/wedge	maximum movement range	repetition; textual parallels
Defense	column of circles	base defense value	number of circles correlates with texts
Weapon/Range	strokes/arrows	attack strength and radius	multiple repetitions on bronze pieces

Section Conclusions

The refined paleographic scheme — in which the **state marker (Marching/Combat)** occupies a fixed position directly beneath the orientation sign — explains many previously debated observations: the near-symmetry of the two sides of the piece, characteristic wear patterns, ease of visual parsing during public games, and the logical organization of turn phases (flipping the piece when switching phases). This convention is one of the defining features of “Kar-Ta-Na,” ensuring its functional stability and its adaptability across diverse cultural contexts.

4.2 Interpretation of Piece Parameters

Below are the results of correlating paleographic data, archaeological contexts, and experimental replication for the primary gameplay parameters encoded on “Kar-Ta-Na” pieces. The descriptions are oriented toward practical reading by a player: what is visible from top to bottom, and how it translates into game values.

Movement

Movement is encoded by a dedicated sign (typically a triangle or wedge) placed beneath the state marker (Marching/Combat). Numerical movement values are usually conveyed by **repetition or modification** of the sign:

- one triangle = movement 1
- two nested triangles = movement 2
- etc.

In some later specimens, a short auxiliary stroke appears beside the triangle — indicating

special movement (e.g., the ability to “jump” over one point).

Experimental replication confirms that this visual encoding is highly readable during play: the triangle is easily recognized at a distance and quickly associated with movement range.

Defense

Defense is represented by a vertical column of circles or dots. The **number of circles** is the unit’s base defense value. Comparative analysis of texts and experimental games shows a direct correlation: the more circles, the more **defense beads** the defender adds to the bag during combat resolution.

Additional defense modifiers are encoded by adjacent signs:

- **plus-dots** indicate a bonus when defending from a fortified position;
 - **small crosses** beside the circles denote reinforced armor (common in later “heavy infantry” pieces).
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Weapon and Range

Weapons are sometimes depicted literally — stylized swords, spears, bows, arrows, oars, etc. The **number of depicted items** typically corresponds to **attack strength**. Examples from the corpus:

- **Three swords** → attack 3
- **Two bows + two arrows** → bow as weapon type, arrows as ammunition; interpreted as **attack 2** with limited shots
- **Bow icon — stroke — four arrows** → read as “bow: attack 2; range 2/4” (first number = base range, second = extended range under special action)
- **Rows of short strokes or arrows** → **range markers**:
 - one stroke = close range (1 point)
 - two strokes = medium range (2 points)
 - arrow with tail = long range (3–4 points)

Some pieces combine weapons: e.g., **sword + thrown weapon** indicates a choice of attack type during the action phase; the number of attack beads depends on the chosen weapon.

Mechanically, the attacker places in the bag a number of beads equal to the sum of all active weapons (e.g., sword 1 + bow 2 = 3 beads), while the defender adds beads equal to defense value plus bonuses for distance and orientation (see §6).

Special Symbols

A set of **special signs** appears on some pieces, encoding non-standard effects. Some are

deciphered, others remain uncertain.

Deciphered Special Signs

- **General’s sign** — stylized crown or warrior figure; grants morale bonuses (in some texts: “removes one defense circle from an adjacent piece when attacking”).
- **Order sign** — small rhombus or double stroke; allows flipping one friendly piece back to marching state or moving an extra point.
- **Fortification sign** — small tower; grants +1 defense on a field marked as fortified.
- **Objective sign** — square with a dot; marks “objective fields,” reaching which may be a victory condition.

Partially Deciphered / Debated Signs

- **Series of dots in a frame** — possibly indicate “patrol” or “logistics” units; in Phoenician sets these often appear alongside economic symbols.
- **Wavy line beneath weapon** — in Cretan sets interpreted as “naval maneuverability,” but unclear in mainland contexts.
- **Small square strokes near orientation** — in some sets grant a flank-attack bonus; in others reduce weapon range; no consensus.

Undeciphered Symbols

Some pictograms remain unresolved, especially those in Hellenistic and Hittite layers, where symbols may have had local administrative or ritual meanings. Several appear to belong to **regional rule modifications** (e.g., tournament conditions or ritual variants), making a unified interpretation impossible with current evidence.

Table of Core Parameters and Their Graphic Markers

Parameter	Graphic Marker	Interpretation
Orientation	arrow/cross	front/flanks/rear
State	triangle / circle or rectangle	marching / combat (two sides of the piece)
Movement	triangle/wedge (repeated)	number of movement points
Defense	column of circles	base defense value
Weapon	swords/bows/arrows (repeated)	attack strength; weapon type
Range	strokes/arrows	attack radius (1–4 points)
Special signs	crown/rhombus/tower	general/order/fortification (varies)

Section Conclusion

The interpretation of piece parameters shows that the encoding system was both flexible and pragmatic: **literal weapon icons** allowed players to assess combat potential at a glance, while **strokes and arrows** provided a compact way to indicate range. The combination of literal imagery (weapons) and abstract markers (defense circles, movement triangles) made “Kar-Ta-Na” suitable for both private play and large public demonstrations.

At the same time, some special signs and local modifications remain undeciphered — a natural outcome for material with such broad geography and long chronology. Certain rule elements were likely oral or regional and thus not consistently recorded in material form. Future finds, comparative analysis of regional sets, and experimental playtests with replicas will help clarify the remaining uncertainties.

4.3 Color and Material Codes

Below are the results of analyzing color and material markers on “Kar-Ta-Na” pieces and their interpretation in terms of **army affiliation**, **piece value**, and **owner’s social status**. The data derive from macro- and microscopic analysis, pigment studies of edges, archaeological contexts, and comparison with textual references.

Table of Codes and Their Interpretation

Attribute	Visual Marker	Interpretation	Archaeological Confirmation
Army affiliation	colored edge; colored inlays	color denotes side/army (usually two contrasting colors)	ochre/pigment residues on edges; color consistency within sets
Piece value	material: clay/stone/bronze/ivory/gold	material encodes value and role (elite units in costly materials)	rarity of bronze/ivory pieces; traces of metal recycling
Owner’s social status	decoration, inlay, warrior/deity imagery	ornate sets indicate elite owners or ritual use	elite pieces in burials and palace contexts
Functional differentiation	color marks on specific symbols	color indicates unit type or special rule	Mari and Nippur texts mention “colored companies”

Army Affiliation

The **colored edge** is the most widespread marker of faction. Many clay and stone tablets preserve traces of ochre, white pigment, or charcoal along the edge; these marks are consistent across all pieces of a set and correspond to textual references to “light” and “dark” sides. Experiments show that a colored edge is easily visible at a distance, explaining its prevalence in public and amphitheater versions of the game.

Piece Value

Material correlates directly with value and function: domestic sets use clay and wood; elite and ritual sets use bronze, ivory, and gold. Archaeological evidence supports this economic logic: valuable pieces appear mainly in burials, temple inventories, and palace contexts. The widespread ancient practice of metal recycling explains why so few complete bronze or gold sets survive — every intact bronze disk is essentially an archaeological miracle.

Owner’s Social Status

Decoration and iconography serve as status markers: pieces with inlay, detailed depictions of warriors or deities, and fine carving are typically associated with elite owners. The Hyppona sets and palace collections from Mari show high artistic quality and clear links to military and administrative elites. Domestic layers, by contrast, contain mostly simple tablets, indicating broad accessibility of the game across social strata.

Functional and Regional Nuances

- **Phoenician and trade-oriented sets** use color and abstract tokens to encode economic roles; color may indicate not an army, but a “resource type” or “tax category.”
 - **Cretan and maritime versions** use color and material to denote fleets and ship types; large inlaid disks served as visual signals for spectators.
 - **Roman standardization** introduced unified color codes and Latinized labels, facilitating interregional play and tournaments.
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Traces of Recycling and Reuse

Archaeological layers show that **many valuable pieces were melted down or repurposed**. This explains the scarcity of bronze and gold sets and highlights that surviving examples come from contexts where recycling was impossible or undesirable (burials, temple deposits, architectural inlays). The presence of artistically valuable pieces in such contexts confirms their role as status symbols.

Limitations and Uncertainties

Despite clear patterns, **regional and chronological variations** exist, and some color codes remain ambiguous (e.g., pigment may have served both decorative and ritual functions). Some sets show mixed markers — color indicating both army and economic role — requiring further contextual analysis.

Section Summary

Color and material codes in “Kar-Ta-Na” served both practical and social functions: **edge color and inlays** enabled rapid identification of sides and unit types, while **material and decoration** signaled piece value and owner status. These codes evolved alongside the game’s spread and adaptation to local cultural needs — from domestic training to public spectacle and ritual representation.

4.4 Hints of a Collectible Nature and a “Deck-Building” Mechanic

A number of contexts and textual fragments provide indirect evidence that “Kar-Ta-Na” sets may not have been static, fixed collections, but instead **assembled and expanded over the course of a player’s life**. In modern terminology, this resembles a **dynamic unit-acquisition system**, in which the owner of a “marching column” gradually increases or modifies its composition through purchase, exchange, or trophies. Below is a synthesis of the main arguments supporting this hypothesis, presented in a neutral scholarly tone.

1. Archaeological Layers with “Mixed” Sets

In several domestic and commercial contexts (especially in Phoenician ports and in the layers of Hyppona), “mixed” sets have been found: clay pieces alongside ivory or bronze ones, with stylistic and workshop markers pointing to different origins and chronological horizons. Such combinations are difficult to explain as single-batch production; a more plausible scenario is that players **augmented their marching columns** with pieces acquired or obtained by other means.

2. Textual Hints of Exchange and Acquisition

Several later texts (commercial records from ports, notes in Ki-Ur-Sag-type “newspapers,” and annotations in the Mari archives) contain formulas that can be interpreted as transfers of “units” between individuals: “three general signs delivered,” “a sword exchanged for two tokens,” etc. The phrasing is not always unambiguous, but in combination with material evidence it suggests a practice of **transferring game pieces outside the context of a single match** — i.e., a market or exchange system.

3. Ownership Marks and Personalization

Some pieces bear personal seals, initials, or small owner graffiti. These marks appear most often on valuable materials (ivory, bronze) and in burials. The presence of personal identifiers indicates that pieces held value as individual objects and could pass between owners through purchase, gifting, or as trophies — becoming part of another player's set.

4. Differences in “Strength” and Rarity of Pieces

Material and iconographic differentiation (ivory pieces with detailed imagery, bronze plates with fine hammering) correlates with gameplay parameters: elite materials often accompany higher attack/defense values or special signs (general, order). This creates an economic logic: owning such pieces increased the “combat value” of a column, incentivizing acquisition and exchange. In practical terms, a strong piece was rare and desirable, and its appearance in a “deck” shifted the balance of power.

5. Trophy and Military-Gift Contexts

Several burials and temple deposits contain pieces clearly originating from foreign cultural spheres (distinct style, foreign workshop stamp). In one case (a coastal officer's burial), a bronze Hittite-style piece lay beside trophy weapons. This suggests that pieces could be transferred as **trophies or diplomatic gifts** — another mechanism for expanding a marching column.

6. Ritual and Commercial Catalogues

Some archival records (including fragments of the Mari “rules”) contain lists of “units” and notes on “deliveries” and “orders” — formulations that can be interpreted as catalogues of available piece types and commissions from workshops. This implies a **market and craft specialization**, where players could order or purchase specific units to strengthen their sets.

7. Tournaments and “Registration” Rules

Some later texts and epigraphic notes (especially in Greco-Roman layers) mention the need to “register” a set before a public match or tournament — listing all pieces and their parameters. Such a practice is typical for systems where set composition varies and where initial conditions must be standardized. This indirectly supports the idea of **dynamic set composition** and the need for regulation in public competitions.

8. The Social Economy of Collecting

Taken together, the evidence suggests that pieces functioned not only as game tools but also

as **objects of exchange, status, and investment**. Owning rare units increased a player's social capital and could signal military or economic standing.

Limitations and Alternative Explanations

It must be emphasized that no direct "instructions for collecting" have been found; all arguments are **indirect** and interpretive. Alternative explanations include: workshop mixing during repairs, ritual deposits from multiple sources, or simple household accumulation without gaming logic. Nevertheless, the combined archaeological, paleographic, and textual evidence makes the hypothesis of **dynamic set-building** plausible and productive for further research.

Section Conclusion

The observed patterns — mixed sets, textual notes on exchanges, personal seals, trophy contexts, and references to set "registration" — form a coherent model in which a player's "marching column" could evolve over time. In scholarly terms, this suggests the existence of a **material-social network of mechanisms for augmenting game sets**, including craft production, trade, trophy practices, and ritual transfer. For historians of games, this resembles a precursor to modern practices of collecting and exchanging game units — but presented strictly within archaeological and textual argumentation rather than contemporary gaming vocabulary.

5. Reconstruction of the Game Board

This section summarizes the archaeological and logical foundations for reconstructing the "Kar-Ta-Na" game board, drawing on material traces (engravings, wall paintings, mosaics) and on the formal requirements of the mechanics (dot-grid topology, connecting lines, piece orientation). The section is divided into two parts: first — empirical evidence; second — the mathematical and functional model of the board.

5.1 Archaeological Traces: Engravings, Wall Drawings, Mosaics

Types of Material Evidence

- **Engraved tiles and stone panels**

Fragments of slabs with carved dots and groove-lines are common in both domestic and public contexts (squares, temples, palaces). These slabs range from small portable boards ($\approx 40 \times 40$ cm) to large stationary monoliths (> 1 m²) intended for public play. The grooves typically connect adjacent dots at right angles, forming the characteristic "node-and-line"

topology.

- **Wall compositions and frescoes**

Several sites (a house in Nippur, a public building in Ubaid, a dwelling in Çatalhöyük) preserve wall-painted dot-grids executed in ochre or charcoal. Nearby impressions of round disks strongly suggest that these were **wall-mounted demonstration boards** — used either for teaching, for spectators, or for ritualized “display games.”

- **Mosaics and marble panels in Hellenistic–Roman contexts**

In later layers, the board is sometimes rendered as a mosaic or an inlaid marble panel with a clear **8×8 structure**. These boards were designed for public spectacles and tournaments, where visibility and durability were essential. The regularity of the grid and the precision of the inlay indicate a high degree of standardization.

- **Floor-tile traces and amphitheater installations**

Large disks and traces of mounting fixtures indicate that some boards were used in amphitheaters and public squares: pieces could be fixed onto stands so that spectators could see their orientation and position from the seating tiers. Abrasive wear on stone floors matches the movement of heavy public pieces.



Archaeological Markers of Board Topology

- **Node-points.**

Most engravings clearly feature **nodes** — the positions where pieces were placed. Nodes often show small depressions or traces of fixation (holes, notches), indicating intentional design for stable placement.

- **Connecting lines.**

Lines between nodes are carved as grooves or thin incised channels. They **always run at right angles**, forming an orthogonal network that links nodes in a grid-like manner. Rare L-shaped or T-junctions appear in local variants, but diagonal links are absent.

- **Positional markers.**

Some boards include “special” nodes — larger, inlaid, or additionally colored — likely representing **objectives**, **fortified positions**, or **scenario-specific points**.

Wear Patterns and Functional Traces

- **Concentration of wear on nodes.**

Micro-abrasion and polishing are consistently concentrated on the node-points themselves, confirming that the game used a **point-based topology**, not a cell-based one.

- **Directional traces.**

Large public boards show linear wear along the connecting grooves, corresponding to frequent movement along these paths. This matches the reconstructed rule that pieces move **along edges**, not across open surfaces.

- **Context of placement.**

Boards in public buildings and squares are often positioned for optimal spectator visibility. This explains the presence of **large disks** and **handles** on pieces — features designed for viewing from amphitheater seating.

5.2 Mathematical Reconstruction of the Grid

Based on archaeological evidence and the logic of gameplay procedures, we propose a formal model of the board suitable for rule reconstruction and experimental play.

5.2.1 Basic Topology: Nodes and Connections

- **The board as a graph.**

The game board can be modeled as a finite undirected graph

[$G = (V, E),$]

where **vertices (V)** are node-points and **edges (E)** are the connecting lines along which pieces move.

- **Vertex degree.**

Each vertex has a degree (d) from 1 to 4, corresponding to connections with 1–4 neighboring nodes at right angles.

Archaeological boards confirm this:

- edge nodes: (d = 1)

- corner nodes: ($d = 2$)
- interior nodes: ($d = 3$) or ($d = 4$)

- **Orthogonality.**

All edges are orthogonal.

Diagonal connections are absent in the standard reconstruction (though some local variants allowed “jumping” moves).

5.2.2 The Standard 8×8 Grid and Its Variants

- **Classical form.**

The most common and most standardized reconstruction is an **8×8 node grid** (eight rows of eight points), forming a 64-node square.

This configuration:

- matches Hellenistic–Roman marble boards,
- is ideal for public play,
- supports clear notational systems.

- **Variability.**

Archaeological evidence shows alternative layouts:

- 7×7, 9×9 grids
- rectangular 8×6 boards
- asymmetrical layouts for special scenarios (naval versions, fortified sectors)

The key requirement is preservation of **local topology** (each node connects to 1–4 neighbors).

- **Reasons for standardization.**

The 8×8 grid balances maneuverability with tactical density.

Its suitability for spectators and for recording matches (tabular notation) likely contributed to its spread.

5.2.3 Connecting Lines and Movement Rules

- **Movement along edges.**

Pieces move along the edges of the graph by a number of nodes equal to their **movement parameter**.

Turns are allowed at nodes; orientation is determined by the piece’s directional marker.

- **Restrictions and special cases.**

Some boards include:

- **barrier nodes** (impassable),
- **fortified nodes** (defense bonuses).

Local rules may have prohibited passing through nodes occupied by stacks, or allowed **jumping** over a single occupied node if the piece bore a special symbol.

5.2.4 Piece Orientation in Space

- **Front/flanks/rear as an orientation vector.**

Each piece has a facing direction — a vector aligned with one of the four orthogonal axes of the board.

This vector determines which adjacent nodes count as **flanks** and **rear**.

- **Visualizing orientation.**

Orientation is read from the arrow-sign on the piece.

When placing a piece, the player aligns the arrow toward the intended front.

- **Mechanical impact.**

Orientation affects combat bonuses:

- flank attacks grant +1 to attack/defense (depending on rules),
- rear attacks grant a larger bonus.

These effects explain the consistent presence of orientation markers on pieces.

5.2.5 Coordinate and Notation System (Practical Recommendation)

For recording and replaying matches, a simple coordinate notation is proposed:

- rows numbered (1–8),
- columns lettered (A–H).

A piece is labeled with a code such as **F-T3**:

- **F** — facing direction,
- **T** — type/identifier,
- **3** — movement/attack value.

A move is recorded as:

F-T3: A2 → A3 (turn right).

This notation is convenient for archiving experimental games and comparing them with ancient “newspaper-style” match records.

Conclusion to Section 5

Archaeological evidence and the formal model of the board converge on the same point: **the “Kar-Ta-Na” board is a point-grid of nodes connected by orthogonal lines**, where pieces are spatially oriented and move along the edges of a graph. The 8×8 standard is the most stable and practical configuration, but local and chronological variations are widespread. Piece orientation (front/flanks/rear) is a key mechanical element, explaining the presence of

corresponding graphemes and the wear patterns observed on the material.

In the next section, we move from the board to a step-by-step reconstruction of gameplay mechanics: turn phases, movement procedures, and combat resolution.

5.3 Board Variants Across Different Periods

Early Bronze Age Boards

Materials and Form

Early boards are most often represented by **portable clay or stone slabs** and wall-painted schemes. Their size ranges from compact pieces ($\approx 40 \times 40$ cm) to large stationary slabs embedded in the floors of public buildings. The topology is a point-grid with connecting grooves; nodes often have small depressions for fixing the pieces.

Topological Features

- **Irregular grids:** 6×6 , 7×7 , and other local variants; node density depends on regional tradition.
- **Special nodes:** marked target points and fortified positions, highlighted by inlay or pigment.
- **Public use:** large stone boards were used in squares and ritual complexes; wall-mounted boards served as demonstration diagrams.

Function and Context

Early boards reflect an experimental stage: strong local adaptation, ritual integration, and domestic use. Boards were often paired with simple clay pieces and a bag of beads as the stochastic element.

Classical Boards of the Phoenician and Hellenistic Periods

Standardization and Aesthetics

In the classical period, we see a clear trend toward **standardization** and decorative craftsmanship. **Marble and inlaid boards** appear, as well as portable boards with a precise regular grid — the most stable form being **8×8 nodes**. Boards become objects of artisanal production and commercial exchange.

Structural Features

- **8×8 as the dominant form:** its convenience for recording matches and for public tournaments makes it the preferred layout in urban centers.
- **Connecting lines:** clean grooves, sometimes filled with contrasting resin or pigment for visibility.
- **Architectural integration:** mosaic boards in houses and public buildings; amphitheater installations with stands for large disks.

Social Role

Boards become part of public culture: tournaments, spectator demonstrations, and official competitions. Rules for registering sets and recording matches appear, requiring field unification and facilitating interregional play.

Late Roman Adaptations

Bureaucracy and Standards

In the Roman era, the board becomes fully standardized and bureaucratized. **8×8** is fixed as the canonical format in most cities; boards are often made of marble or bronze panels and bear Latinized labels. Official regulations for public tournaments and match records emerge.

Technical Innovations

- **Piece fixation:** holes and stands for large disks to ensure visibility from the stands.
- **Notation and registration:** formal lists of sets before tournaments; standardized node and field labels.
- **Mobility:** folding wooden boards and travel sets for itinerant officer-players.

Cultural Consequences

The Roman version of the board serves simultaneously as entertainment, military demonstration, and status symbol. Standardization facilitates trade in sets and the development of specialized workshops, but economic logic leads to the melting down of valuable materials — hence the rarity of complete elite sets.

Section Summary

The evolution of “Kar-Ta-Na” boards progresses from the local, variable topologies of the Early Bronze Age to the standardized, public-oriented forms of the Classical and Roman periods. The **8×8** grid emerges as an effective compromise between maneuverability and spectacle, yet regional and chronological variants remain significant: they reflect local tactics, ritual practices, and the social functions of the game.

6. Reconstruction of Game Mechanics

6.1 The Marching Column as the Prototype of a “Deck”

Brief Definition

The **marching column** is an ordered collection of a player’s units, stored and expanded outside the board; it functions as a dynamic reserve from which pieces are introduced into the match. The marching column forms historically and socially: some units are inherited, some acquired, some taken as trophies.

1. Structure of the Marching Column

- **Ordered sequence** — the column is a stack or list in which each unit occupies a fixed position; order matters for deployment mechanics.
 - **Layers and “life stacks”** — within the column, certain unit types may be stored as stacks (multiple identical tablets = the “lives” of a single combat unit).
 - **Visibility hierarchy** — the top unit of the column is usually **open** and visible to the opponent; lower units remain hidden or only partially known.
-

2. Order of Deployment onto the Board

- **Turn-based deployment rule** — during the “deployment phase,” a player may place the top piece of their column onto the board or perform a substitution by removing the top piece and revealing the next one.
 - **Deployment constraints** — deployment may require conditions: an available edge node, correct orientation, or payment of a resource (texts mention “grain,” “token”).
 - **Tactical significance of order** — the order of the column shapes strategy: deploying the top piece reveals part of one’s plan; keeping a strong unit deeper in the column conceals reserves.
 - **“Transition” mechanic** — when deploying a piece, only the top element is removed; if it is a life-stack, only the top tablet of the stack is taken.
-

3. Open Information About the Top Piece

- **Transparency principle** — the top piece of the column is displayed to both sides, providing public information about the player’s immediate options.
- **Functions of openness:**
 - **reducing uncertainty:** the opponent knows what unit may appear next;
 - **social signaling:** displaying a powerful top piece could serve as a status gesture or challenge;

- **tournament regulation:** later texts mention “lists of top units” during set registration.
 - **Visibility variants:**
 - **Full openness** — the entire top piece is visible;
 - **Partial openness** — only the state marker or orientation is visible;
 - **Hidden top piece** — a rare local rule where deployment reveals the piece randomly.
-

4. Gameplay Consequences of Order and Openness

- **Planning and bluffing** — an open top piece enables strategic planning and counterplay; at the same time, bluffing is possible through the arrangement of deeper layers.
 - **Economy of reinforcement** — since elite units are rare, the column’s order reflects investment: “working” units occupy upper slots, valuable ones are stored deeper.
 - **Trophies and exchanges** — captured pieces could enter the victor’s column, altering its strength and order; texts record cases like “transfer of three general signs” as reinforcement.
-

5. Practical Formalization for Experimental Reconstruction

- **Recommended prototype column:** a model of 12–20 slots; the top slot is open; slots 1–4 are immediately deployable; slots 5–12 form the reserve.
 - **Deployment rule:** during the deployment phase, a player may place the top piece on any free edge node if its movement ≥ 1 ; alternatively, spend one “logistics token” to deploy it inside the board.
 - **Stacking rule:** identical tablets are stored as a stack; when taking damage, only the top tablet is removed.
 - **Tournament regulation:** before a match, players present the top three pieces of their columns for recording; any later reinforcements are added to the protocol.
-

6. Limitations and Open Questions

- **Ambiguity of sources** — texts provide fragmentary hints about exchange and registration, but no direct “instructions for assembling a column”; the proposed formalizations are working hypotheses for field testing.
- **Local variations** — in some regions, order may have been less rigid: trophy inclusion, workshop commissions, and ritual transfers created diverse column structures.
- **Further verification** — experimental matches with replicas and analysis of archaeological traces (seals, owner graffiti) will help determine how widely this model was used.

Summary

The marching column is a functional ancestor of the modern concept of an ordered reserve: its structure, deployment order, and open top piece created a balance between strategic predictability and the ability to accumulate strength through exchanges and trophies. Formalizing these mechanisms provides a working foundation for experimental reconstruction and for correlating gameplay with textual and material evidence.

6.2 The Two Sides of the Piece: Movement and Actions

The fundamental construction of a piece in the reconstructed “Kar-Ta-Na” model is a **double-sided tablet**, whose two faces are nearly identical in layout but differ in a single key element: the **state marker**. One side shows the **Marching State** (triangle), the other the **Combat State** (rectangle or circle).

This simple flip mechanic establishes the core rhythm of the game: switching from movement to action and back again.

Functional Logic

Two phases embodied in a single object

- **Marching State**

The piece is optimized for movement:

- maximum **movement** value,
- reduced combat parameters (attack/defense),
- ability to “leap” or reach distant nodes.

- **Combat State**

The piece is optimized for interaction:

- increased **attack** and/or **defense**,
- access to special actions (ranged fire, assault, command),
- reduced movement.

Flipping the piece between these states is a key tactical decision — the equivalent of switching a military unit between marching formation and battle formation.

Flip Mechanics

- **Flip as an action**

Changing state usually requires spending the action phase or a special resource; in

some local variants, flipping is free when deploying from the column.

- **Flip restrictions**

Flipping may be forbidden under “panic” conditions or when occupying a fortified node. Some texts mention a “delay” after flipping — the piece cannot act on the same turn it was flipped.

- **Information clarity**

The upper zone of the piece (orientation + state marker) is instantly readable, making state changes clear to both players and to spectators in public matches.

- **Interaction with other parameters**

The state affects how subsequent parameters are interpreted: **movement** → **defense** → **weapon/range**.

For example, the same weapon icon may mean “carried weapon” in marching state but “active attack” with a defined range in combat state.

Analogies with Military Maneuvers

March and Deployment

Marching state corresponds to a column on the move: high mobility, vulnerability on contact. Switching to combat state is **deployment into battle formation**, analogous to real-world tactics.

As in actual warfare, deployment requires time and space — represented in the game by spending a turn or requiring a free node.

Flanking and Orientation

Orientation defines front, flanks, and rear.

Flank or rear attacks grant bonuses — a direct analogy to historical maneuvers: outflanking, breaking formation, rear assaults.

The mechanics reward maneuver: marching to outflank, then flipping to combat state for the strike.

Rotation and Fire Discipline

The combination of marching movement and subsequent combat flipping models “approach-strike” tactics:

- rapid advance (march),
- deployment and attack (combat).

For ranged units, this may appear as “firing on the move” with reduced efficiency, or deploying for accurate long-range fire.

Reserves and Rotation

The marching column and stacked layers model reserve rotation: elite units are kept in reserve and deployed at decisive moments — mirroring historical practice.

Fortifications and Positional Warfare

Combat state in a fortified node grants defensive bonuses, reflecting the difference between maneuver warfare and positional defense.

Practical Rules for Experimental Reconstruction

Flipping

- **Rule:** flipping from marching to combat state requires an action; flipping from combat to marching is free during the movement phase.
- **Consequence:** a piece flipped into combat state during the action phase cannot move until the next turn.

Parameter Changes

- **Marching State:**
movement ($= M$), attack ($= A_m$), defense ($= D_m$)
- **Combat State:**
movement ($= M - 1$) or $(M/2)$,
attack ($= A_b$) (typically $A_b > A_m$),
defense ($= D_b$) (typically $D_b > D_m$)

Example:

A piece with base values ($M = 3$, $A_m = 1$, $D_m = 1$) may have in combat state:
($M = 1$, $A_b = 3$, $D_b = 2$).

Range Restrictions

- Weapon range is adjusted by state:
 - marching state: range reduced by 1,
 - combat state: full range.

This models the need to deploy for effective ranged fire.

Tactical Techniques

- **Recon and flanking:** use marching state to maneuver around flanks, then flip to combat for the strike.
 - **Holding a node:** flip to combat state on a fortified node for a defense bonus.
 - **Reserve rotation:** deploy from the column in marching state, reposition quickly, then flip to combat at the critical point.
-

Limitations and Empirical Testing

- **Local variations** — texts and finds indicate many regional rules: in some areas flipping was free, in others it required a resource. Reconstruction must remain flexible and tested through experimental play.
 - **Ambiguous symbols** — some pictograms affecting flipping remain undeciphered; further correlation with texts and artifacts is needed.
 - **Spectacle factor** — in public versions, flipping had a visual function: the dramatic reveal of the other side may have influenced rules (e.g., mandatory visible flipping).
-

Summary

The double-sided nature of the piece is a simple yet powerful mechanic that unifies movement and interaction into a coherent tactical system.

Flipping between **marching** and **combat** states models real military operations: marching, deployment, striking, holding ground, and rotating reserves.

This design makes “Kar-Ta-Na” both an effective tool for teaching military logic and a compelling public spectacle.

Experimental matches with replicas will help refine numerical parameters and local rule variations.

6.3 The Movement Phase

The movement phase is the structured portion of the turn in which the maneuverability of columns is expressed, positions are shifted, and preparations for combat interactions are made. Below is a practical reconstruction of procedures and rules derived from archaeological evidence, paleography, and experimental replication.

Movement Along the Lines

Topology of Movement

The board is treated as a graph of nodes and edges; pieces move **along the edges** from node to node. Movement is measured in nodes and determined by the piece’s **movement parameter** (denoted (M)).

Basic Movement Rule

- **Each piece may move a number of nodes equal to its (M)** during the movement phase.
- Turning at nodes is allowed and does not cost additional distance unless specified by local rules.
- Passing through a node occupied by one’s own stack is allowed; passing through a

node occupied by an enemy stack is forbidden (entering such a node triggers combat resolution in the action phase).

Restrictions and Obstacles

- Barrier nodes and fortified nodes are marked on the board; passing through them is either forbidden or requires an additional cost (e.g., reducing (M) by 1).
- Life-stacks occupy a node completely; jumping over an occupied node requires a special “jump” symbol or corresponding ability.

Special Movement Types

- **Jumping:** allows a piece to leap over one occupied node; encoded by an extra stroke next to the movement triangle.
 - **Sliding march:** when moving along a continuous straight line without turns, certain unit types (naval versions, chariots) gain +1 movement.
-

Turning

Definition

A **turn** is a change in the piece’s orientation (front vector) at a node. Orientation determines flanks and rear and affects combat modifiers.

Variants of Turning

- **Turn as part of movement:** turning is free and may be performed at any node within (M). Typical of fast, maneuver-oriented systems.
- **Turn as an action:** turning requires the action phase or a special resource; used in “heavier” local variants where deployment takes time.
- **Restricted turning:** turning allowed only on nodes with degree ($d \geq 2$) (not on the board edge) or only if an adjacent node is free.

Tactical Consequences

- Turning in combat state often prevents further movement in the same turn.
 - Turning in marching state enhances maneuverability and enables rapid flanking.
 - Flank/rear attacks are determined by the relative orientation of attacker and defender at the moment of combat resolution.
-

Deploying Pieces from the Column

Deployment Mechanics

- **Deployment** is the act of placing the top piece of the marching column onto the board. Deployment occurs either during the movement phase or in a separate deployment phase depending on local convention.
- **Standard rule:** the top piece may be placed on any free edge node, oriented toward the board; deployment counts as part of the turn (reducing (M) or replacing movement).

Conditions and Restrictions

- A free edge node is required; in some variants, deployment into the interior is possible for a resource cost (logistics token).
- Deployment in combat state may be forbidden; deployment in marching state is preferred.
- If the top piece is a life-stack, only the top tablet is placed; the unit is considered present on the board until the stack is exhausted.

Information at Deployment

- **The top piece is open:** the opponent sees its parameters and orientation before deployment. This provides strategic information and reduces uncertainty.
- In tournaments and public matches, registration of the top N pieces of the column was often required before the game.

Interaction Between Deployment and Engagement

- Deploying onto a node occupied by an enemy triggers immediate conflict resolution (in the action phase) or is forbidden depending on local rules.
- Deploying adjacent to an enemy may trigger "reaction fire" by the defender (a local option).

Examples of Typical Moves

- **Example 1:** a piece with (M = 3) in marching state moves 3 nodes along a line, turns at a node (turn is free), and remains in marching state for the next turn.
- **Example 2:** the top piece of the column is deployed onto an edge node; the opponent immediately flips it into combat state using their action, preparing it for attack next turn.
- **Example 3:** a piece with a jump symbol leaps over an occupied node and lands on the next one, but loses 1 point of (M) at the end of the turn.

Recommendations for Experimental Rule Testing

Basic Test Protocol

- Use an 8×8 board; standard (M) values in replicas: light infantry (M = 3), heavy infantry (M = 1), cavalry (M = 4).
- Test three turning variants: free, action-based, restricted. Compare dynamics and balance.
- Deployment: allow free deployment on edge nodes; in an alternative variant, allow interior deployment for a logistics token.

Evaluation Metrics

- **Maneuverability:** average number of nodes traversed by a piece over 3 turns.
- **Deployment frequency:** how often pieces are flipped into combat state.
- **Outcome influence:** correlation between column order and victory.

Summary of Section 6.3

The movement phase is the tactical core of the game: movement along lines, orientation control, and effective deployment from the column form the foundation of maneuver strategy.

The division between marching and combat states, along with turning and deployment rules, creates a rich tactical palette that must be tested experimentally with replicas under different parameter sets.

6.4 The Action Phase

The action phase is the moment when the maneuvers prepared during the movement phase turn into direct interactions: attacks, special abilities, and administrative operations (orders, fortifications, etc.).

What follows is a working reconstruction of procedures and rules derived from the material corpus, textual hints, and experimental replication. Everything here is reconstruction rather than canon; many details remain subject to testing.

Attacks

General Logic

An attack is an attempt by one piece (or a group of pieces) to change the state of another piece/stack on the board: remove a “life,” push it back, capture a node, or alter the opponent’s orientation.

Attacks are always initiated during the action phase and depend on three components encoded on the piece: **weapon (attack strength)**, **range**, and **the defender’s protection**.

Reconstructed Attack Resolution Sequence

1. **Declaration** — the attacker declares the target and the type of attack (melee, ranged, assault, thrown weapon, etc.).
2. **Range and orientation check** — verify whether the target is within weapon range and whether it benefits from fortification; the target's orientation (flank/rear) provides modifiers.
3. **Parameter comparison** — compare numerical values: the attacker's **attack** and the defender's **defense**, including modifiers (flank +1, rear +2, fortification -1, etc.).
4. **Stochastic resolution** — when values are equal or the outcome is not obvious, a stochastic mechanism is used (a bag of beads/grains). Several texts explicitly mention "light" and "dark" grains. The reconstruction uses the following working algorithm:

- **Bag mechanic (working hypothesis):**

The attacker and defender each place into the bag a number of beads equal to their modified combat values: attacker places (A) beads of their color, defender places (D) beads of theirs.

One bead is drawn.

If the bead is the attacker's color — the attack succeeds; if the defender's — the attack is repelled.

- **Interpretation:**

This creates a simple probabilistic relationship: the stronger side has a higher chance of success, but outcomes are not deterministic.

Some texts suggest alternative methods (multiple draws, majority count), so the bag mechanic should be treated as a working model, not a final rule.

5. **Consequences of success/failure**

- **Success:** the top plate of the target's life-stack is removed, or the piece is eliminated; additional effects may occur (pushback to an adjacent node, capture of the node).
- **Failure:** the attacker remains in place, may suffer counter-damage (depending on local rules), or may be forced back into marching state.
- **Critical effects:** certain symbol combinations (general + assault) may trigger enhanced outcomes — removing two "lives," capturing a stack, etc. These appear on elite pieces and in specific texts.

Example

Piece A: attack 3 (three swords), flank bonus +1.

Piece B: defense 2 (two dots).

After modifiers: attack = 4, defense = 2.

The bag contains 4 attacker beads and 2 defender beads; one is drawn — probability of success = (4/6).

Special Actions

Classification

Special actions are a set of locally varying abilities encoded on pieces by special symbols. They fall into several functional groups:

Command Actions (general symbol, order symbol)

- **Order:** allows flipping or moving an adjacent piece, granting an extra action, or canceling an opponent's flip.
 - **Morale:** temporary increase of attack/defense for adjacent pieces.
-

Ranged and Siege Actions (bows, slings, naval artillery)

- **Shot:** a ranged attack without direct contact; effectiveness depends on state (march/combat) and cover.
 - **Siege strike:** with a siege symbol, a piece may attack a fortified node with reduced efficiency but with a chance to break fortification.
-

Logistical and Economic Actions (Phoenician variants)

- **Trade operation:** exchange of tokens, gaining a resource, temporary reinforcement of the column.
 - **Repair:** restoring one "life" of a stack at the cost of a resource.
-

Manipulative Actions

- **Interception:** ability to interrupt an opponent's deployment under certain conditions.
 - **Capture:** on a successful attack, instead of destroying the piece, the victor may add it to their column as a trophy.
-

Costs and Restrictions

Special actions often require expenditure:

- spending a turn,
- removing the top plate of a stack,
- paying a logistics token,
- meeting a prerequisite (e.g., occupying a fortified node).

Some texts mention "light grains" as the cost for issuing an order — possibly representing resource expenditure.

Ambiguous and Undeciphered Symbols

Some special symbols are rare and their function remains unclear: framed dot series, wavy lines, small square strokes.

These likely encoded local abilities (patrol, scouting, diplomacy) and require further study.

Flipping Pieces During the Action Phase

Role of Flipping

Flipping is a key tool: it changes the piece's state (march $\leftrightarrow$ combat) and thus its parameters. In the action phase, flipping is used for:

- **Activating combat capabilities** — flipping into combat state grants access to full attacks and special actions.
 - **Marking resource expenditure** — flipping may indicate that a unit has been “used” this turn; in some variants, a flipped piece cannot act again until the next turn.
 - **Representing damage** — in some local rules, flipping the top plate of a stack indicates damage (the plate is flipped first, then removed on the next successful hit).
-

Reconstructed Flip Rules

- **When flipping is allowed:** flipping into combat state is usually done during the action phase as preparation for attack; flipping into marching state is usually done during the movement phase.
 - **Cost of flipping:** in the base reconstruction, flipping into combat state requires an action; in lighter variants, it is free when deploying from the column.
 - **Restrictions:**
 - a piece flipped into combat state during the action phase cannot move until the next turn;
 - a piece flipped into marching state loses part of its attack/range for the remainder of the turn.
 - **Visual and social function:** flipping is a visible act, important in public matches; it also served as a signaling practice (display of strength, challenge).
-

Example Scenario

A player deploys a piece in marching state onto an edge node (movement phase).

In the action phase, they flip it into combat state (spending an action), declare a ranged attack at distance 3, check range, place beads into the bag, and resolve the outcome.

Conclusion to Section 6.4

The action phase is the center of “Kar-Ta-Na’s” conflict dynamics: it combines quantitative

parameters (attack, defense, range), a stochastic element (the bead bag), and qualitative abilities (orders, fortifications, trophies).

Flipping between marching and combat states is a simple yet expressive mechanic linking maneuver and interaction.

Many details (costs of special actions, exact modifier formulas, rare symbols) remain hypothetical and require experimental testing with replicas and further comparison with textual sources.

6.5 The Randomness System

Below is a working reconstruction of the stochastic mechanism which, based on the combined material and textual hints, most plausibly underlay combat resolution in "Kar-Ta-Na."

It describes the bead-bag device, a simple probabilistic model, and a practical combat formula incorporating modifiers (flank, rear, fortification, distance).

All formulas presented here are reconstructions suitable for experimental testing.

The Bag of Black and White Beads

Mechanism Structure

- **Materials:** a bag containing small beads or grains of two contrasting colors (texts and finds mention "light" and "dark" grains).
- **Procedure:** during combat resolution, the attacker places into the bag a number of beads of their color equal to their combat value; the defender places beads of their color equal to their defense value.

One bead (or several, depending on local rules) is drawn.

The color of the drawn bead determines the outcome: attacker's color = success; defender's color = defense holds.

Practical Notes

- The bag provides a simple, intuitive stochastic system: the greater a side's relative strength, the higher its chance of success.
- Some texts mention variants with multiple draws or majority counting; these produce smoother probability curves and allow modeling of multiple damage outcomes.

Probabilistic Model

Basic Single-Draw Model

Let the attacker's strength be (a), and the defender's strength be (d).

The bag contains (a) attacker beads and (d) defender beads.

The probability that a single draw yields an attacker bead is:

$$[P(\text{success}) = \frac{a}{a + d}.]$$

Correspondingly, the probability of successful defense is:

$$[P(\text{defense}) = \frac{d}{a + d}.]$$

Modifiers

Modifiers (flank, rear, fortification, distance, general, etc.) are introduced as additive adjustments:

$$[a' = a + \Delta_a, \quad d' = d + \Delta_d,]$$

where Δ_a and Δ_d are the total modifiers.

Then:

$$[P(\text{success}) = \frac{a'}{a' + d'}.]$$

Alternative Resolution Methods

- **Multiple draws:** draw (k) beads and decide by majority; this produces a sharper dependence on the ratio (a : d).
- **Counting successes:** attacker places (a) beads, defender (d); draw (n) beads and count the number of attacker beads (X).
(X) follows a hypergeometric distribution, allowing damage to scale with the number of attacker beads drawn.
This variant is useful for modeling multi-life stacks.

Reconstructed Combat Formula

Basic One-Step Formula

Using the “one draw = one resolution” procedure:

On success, the attacker removes one “life” (top plate of the stack); on failure, the attack is repelled.

Thus, the probability of removing one life is:

$$[P_{\text{remove}} = \frac{a'}{a' + d'}.]$$

Expected Number of Lives Removed in (n) Independent Attacks

If the attacker performs (n) attacks with identical parameters (or a group attacks simultaneously with separate resolutions):

$$[\mathbb{E}[\text{losses}] = n \cdot \frac{a'}{a' + d'}.]$$

Example

Attacker: three swords $\rightarrow (a = 3)$.

Defender: two defense dots $\rightarrow (d = 2)$.

$$[P_{\{\text{success}\}} = \frac{3}{3 + 2} = \frac{3}{5} = 0.6.]$$

If the attack is from the rear and rear grants +2:

$$[a' = 5, \quad d' = 2,]$$

$$[P_{\{\text{success}\}} = \frac{5}{7} \approx 0.714.]$$

Distance and Weapons

Distance is encoded by strokes or arrows.

Practical reconstruction: at maximum range, apply a distance penalty ($\Delta_{\{\text{dist}\}}$) (typically -1 or -2).

Example: a bow with base attack 2 and max range 4, firing at distance 4, may have effective attack ($a' = 2 + \Delta_a - 1$).

The corpus example “bow attack 2, range 2/4” is interpreted as:

- base attack 2,
- effective range 2 for standard fire,
- up to 4 for a deployed action with reduced efficiency.

Group Attacks and Summed Strength

When multiple pieces attack a single target simultaneously, their attack values sum:

$$[A = \sum a_i.]$$

Then:

$$[P_{\{\text{success}\}} = \frac{A'}{A' + d'}.]$$

This naturally incentivizes coordination and encirclement.

Practical Recommendations for Experimental Testing

- **Basic protocol:**

Use a bag with ~20 beads of two colors; represent each unit's strength by the number of beads added.

Test three resolution variants:

- single draw,
- three draws with majority,

- drawing (n) beads and counting attacker beads.

- **Modifier parameters:**

Working values: flank +1, rear +2, fortification –1, distance penalty –1 at maximum range.

These can be adjusted for balance.

- **Metrics:**

- empirical success probability vs. theoretical ($a'/(a' + d')$),
- average number of turns required to remove a stack of (k) lives.

Limitations and Notes

- **Indirect evidence:** archaeological and textual data indicate the presence of a bag and the idea of “number versus number,” but the exact procedure (single draw, multiple draws, counting) remains uncertain.
- **Local variants:** different regions and periods may have used different stochastic methods; the proposed model is a testable working hypothesis.
- **Social function of randomness:** the bead draw was not only a mathematical device but also a ritual and theatrical act — the reveal of the bead likely had symbolic significance.

Summary of Section 6.5

The bag of black and white beads is a simple, robust, and archaeologically plausible stochastic mechanism.

The basic probability formula

$$[P = \frac{a'}{a' + d'}]$$

provides an intuitive and easily reproducible model for experimental reconstruction of combat in “Kar-Ta-Na.”

Variants with multiple draws and success counting allow modeling of more complex effects and multi-life damage.

The proposed working parameters and protocols are suitable for field testing with replicas and for further calibration against archaeological data.

7. Social and Cultural Context of the Game

7.1 The Role of the Game in Bronze Age Society

Below is a synthesis of archaeological, paleographic, and contextual observations regarding the social functions performed by **Kar-Ta-Na** in Bronze Age communities.

Two major spheres of influence emerge: **military training** and **ritual functions**, often overlapping and mutually reinforcing.

Military Training

A Tactical Training Model

Kar-Ta-Na functioned as a compact model of battlefield interactions: maneuvers, deployment, flanking attacks, and the use of reserves could be practiced in a safe, representative format.

The game allowed officers and junior commanders to rehearse decision-making, distance assessment, and coordination of multiple units without real-world risk.

Development of Command Logic

Concepts such as the **marching column**, deployment order, reserve rotation, and piece orientation modeled military organization and command procedures.

This fostered skills in planning, resource allocation, and risk management.

Educational Function

The game served as a tool for transmitting tactical knowledge across generations: younger warriors learned through examples, while elders demonstrated techniques.

The presence of textual "rules" and recorded matches in palace archives indicates formal instructional use.

Psychological and Moral Preparation

Tournaments and public matches acted as rehearsals for moral trials: demonstrating courage, resilience, and the ability to make risky decisions.

Victory in a public match could enhance a commander's reputation and authority.

Logistics and the Economics of War

Mechanics of column replenishment, trophy integration, and commissioned pieces reflect the economic aspects of military activity: supply, exchange, and distribution of rare resources (bronze, ivory) parallel real logistical systems.

Ritual Functions

Funerary and Cult Contexts

The presence of pieces in burials and temple deposits indicates their role as ritual objects: sets could accompany the deceased as a "marching column" into the afterlife or serve as

symbols of status and professional identity.

Divination and Symbolic Modeling of Fate

The stochastic element (the bead bag) and the public act of drawing a bead carried ritual meaning: the result could be interpreted as divine will or fate, making the game a tool for divination and collective decision-making in sacred contexts.

Ritualized Tournaments and Status Display

Public matches—especially those using elite materials and inlays—served a representational function: displaying wealth, political influence, and cultural competence.

Awarding winners, gifting pieces as diplomatic tokens, and incorporating trophies into rituals enhanced the game's social significance.

Integration with Religious Symbolism

Iconography of gods and sacred motifs on elite pieces indicates syncretism: game units could function both as modeling tools and as cult objects, especially in contexts where warfare and divine patronage were closely linked.

Intersections of Military and Ritual Spheres

Dual Function of Sets

Many sets served simultaneously as training tools and ritual artifacts: the same pieces could be used in drills, then offered to a temple or placed in a burial.

This duality strengthened the cultural resilience of the practice.

Social Inclusion and Stratification

Simple clay sets made the game accessible to broad social strata; elite bronze and ivory sets served as status markers.

The game functioned as a field of both social competition and integration.

Summary of Section 7.1

Kar-Ta-Na was not merely entertainment: it was a multi-layered technology for modeling social and military processes.

It performed practical functions of training and education while simultaneously serving ritual and representational purposes.

This dual nature explains the game's wide distribution, adaptability to local contexts, and persistence in the material culture of the Bronze Age.

7.2 Spread During the Hellenistic Period

Short answer:

During the Hellenistic era, Kar-Ta-Na spread along maritime and overland trade corridors, adapting in colonies through local workshops, public tournaments, and diplomatic exchanges; this process produced standardized boards and regional rule variants.

Trade Routes as Vectors of Transmission

Maritime Corridors of the Eastern Mediterranean

Phoenicia, Cyprus, and the southern coast of Asia Minor served as major arteries for the transmission of material culture and game sets.

Sets produced in one center traveled to colonies and ports alongside goods and craftsmen.

Overland Trade Networks

Routes through Antioch, the Tigris–Euphrates corridor, and Greco-Roman commercial roads facilitated the spread of textual rule summaries and “catalogues” of piece types between administrative centers.

Mechanisms of Adaptation in Colonies

Craft Localization

Workshops in colonies quickly adopted the form and symbolism of the pieces but used local materials (marble, ceramics, bronze) and local iconographic motifs.

This explains the emergence of regional variants while preserving a shared “lexicon” of signs.

Social Integration

In colonies, the game entered the repertoire of public events: tournaments, ritual demonstrations, and diplomatic displays.

This encouraged the standardization of boards (often 8×8) and rules for spectator-friendly matches.

Commercialization of Sets

Catalogue entries and workshop orders in archives indicate a market for sets: players could commission “elite” pieces from local artisans or exchange them through trade networks.

Local Transformations of Rules and Forms

Colonies adapted the game to their needs:

- maritime communities emphasized naval themes (ships, sails, range),
- trading centers introduced economic modifiers (exchange, repair),
- military centers expanded command symbols.

These transformations explain why the archaeological corpus contains both standardized elements and unique local symbols.

Archaeological and Textual Indicators of Spread

- **Similar sets in ports and colonies** (matching color codes, identical orientation signs) indicate direct transmission of sets.
 - **Records of deliveries and “orders”** in later archives confirm commercial activity and craft specialization.
-

Summary

The Hellenistic period transformed local gaming practices into a **trans-regional cultural technology**:

trade routes and colonies became channels of transmission, workshops became centers of adaptation, and public/ritual contexts became arenas of standardization and innovation.

Targeted archaeological comparisons of finds from port and colonial layers can further refine this picture.

7.3 Decline and Disappearance in the Roman Era

Overview of the Process

During the period of Roman expansion and cultural integration, the material and textual footprint of **Kar-Ta-Na** gradually diminishes.

This was not a sudden collapse but a multifactorial process: competition with new gaming practices, institutional standardization of entertainment, and transformations in military culture eroded the game’s former social niche.

Competition with Roman Games

New Preferences and Mass Culture

The Roman era introduced a wide range of popular board and gambling games (dice games, *tabula*, *ludus latrunculorum* variants), which were easier to mass-produce and more readily standardized for public spectacles and tabular tournaments.

These games required less artisanal specialization and fit more naturally into commercial markets, making them more accessible and popular.

Institutionalization of Spectacle

Roman public culture strengthened the role of amphitheaters, baths, and forums as entertainment centers.

Games that adapted easily to mass formats and commercial rules displaced more specialized local practices.

Where **Kar-Ta-Na** required complex sets and artisanal support, Roman analogues offered simplicity and standardized regulation.

Economic Factors

Production and trade of Roman game sets operated on a larger commercial scale.

Workshops oriented toward mass demand displaced local artisans who produced elite **Kar-Ta-Na** pieces.

Bronze melting and ivory repurposing in Roman economic conditions further reduced the number of surviving sets.

Loss of Military Symbolism and Functional Transformation

Shift in Military Paradigm

The Roman army and its ideology differed from many Bronze Age models: centralization, standardized legionary tactics, and bureaucratized command reduced the need for local tactical “schools” in the form of tabletop rehearsal.

Where the game once served as a model for maneuver and command training, the Roman system relied on other forms of instruction (field drills, written manuals, centralized logistics).

Decontextualization of Symbolism

The military semantics of the pieces—orientation, flanks, rear, generals—lost practical relevance under new tactical and administrative norms.

As real military procedures became more formalized, the symbolic language of the game no longer matched contemporary models, reducing its utility for professionals.

Ritualization Instead of Practice

In some regions, **Kar-Ta-Na** underwent transformation: from a training tool it became a ritual or collectible object.

Such transformed sets appear in burials and temple deposits, but active gameplay gradually faded.

Social and Material Consequences of Decline

Reduction of the Craft Network

Decreased demand led to the disappearance of specialized workshops and to the melting down of valuable materials.

Elite pieces became rare, while mass-produced Roman sets became the norm.

Local Survival

In certain peripheral regions and religious contexts, elements of **Kar-Ta-Na** survived longer: local traditions, ritual practices, and collecting habits preserved isolated variants for several centuries.

Archaeological “Gap”

The combination of melting, repurposing, and shifting cultural priorities explains why the material corpus of **Kar-Ta-Na** shrinks sharply in the Roman era and why many late references are fragmentary and difficult to interpret.

Conclusion to Section 7.3

The decline of **Kar-Ta-Na** in the Roman era resulted from cultural competition, economic reorganization, and transformations in military practice.

The game did not vanish instantly: some elements were integrated into new forms, some sets moved into ritual and collectible spheres, while the mass gaming scene was overtaken by more standardized Roman practices.

This explains both the gradual disappearance of gameplay traditions and the survival of isolated artifacts and motifs in later archaeological layers.

8. Written Tradition Sources

Below is a synthesis of key written testimonies, legends, and administrative notes that illuminate the semantics, functions, and spread of **Kar-Ta-Na**.

The texts range from mythological narratives to economic records; together they form a multilayered corpus that must be read alongside archaeology.

8.1 Mesopotamian Records

Nature of the Corpus

Mesopotamian sources include mythological legends, administrative notes, and fragments of “rules” in palace archives.

The texts often link the game to divine knowledge and military planning.

Illustrative Fragments

Ki-Ur-Sag — Field of Heroes (Nippur)

“When the gods were still dividing the fates of cities, Enki laid out before Enlil a field of lapis and bitumen. He placed upon it the figures of heroes — *ur-sag* — and moved them without speaking...”

Šaṭṭu Šābi — The Army’s Path (Babylon)

“King Sarru-kin ordered the scribes to make a board so that the army might march across it before stepping onto the enemy’s land.”

Key Conclusions

- **Dual function:** texts attribute both sacred (priestly “tablets of fates”) and practical (military modeling) roles to the game.
- **Archival traces:** mentions of “tens and hundreds” and “transfer of signs” align with the idea of parameterized pieces and unit exchange.
- **Regulation:** formulas and accounting notes indicate institutional use in temples and palaces.

8.2 Egyptian Inscriptions with Game-Related Terms

Nature of the Corpus

Egyptian texts more often associate the game with royal and priestly practice, with concepts of *Maat* and the verification of order before a campaign.

Illustrative Fragment

Ta-mesher-netjeru — Field of Divine Order (Memphis)

“Thoth drew a field on the temple floor and arranged the figures of the armies... This must be in the hands of pharaohs before they take the scepter.”

Key Conclusions

- **Ritual verification:** the game served as a symbolic test of order and legitimacy.
- **Royal context:** texts indicate elite use in royal and priestly circles, consistent with elite sets found in burials and temples.
- **Local semantics:** the Egyptian version emphasizes “proper order” (*Maat*), which may

have influenced rules and interpretation of special signs.

8.3 Phoenician and Hellenistic Texts

Nature of the Corpus

Phoenician records and Hellenistic catalogues reflect commercial and maritime adaptation: mentions of orders, deliveries, and simplified abstract pieces.

Illustrative Fragment

Derekh Šaba' — The Army's Way (Tyre)

"Our fathers played to measure risk. We continue — for the same reason."

Key Conclusions

- **Commercial logic:** Phoenicians were the first to adopt abstract pieces and market-based trade in sets.
- **Catalogues and orders:** archaeological and textual evidence points to workshops and the ability to commission specific piece types.
- **Maritime adaptation:** Hellenistic layers show naval variants and standardized boards (often 8×8) for public tournaments.

8.4 Later Mentions in Greek and Latin Texts

Nature of the Corpus

Greek and Latin sources document the game's transformation within classical and Roman culture: from Spartan educational practice to the bureaucratized Roman *Ludus Ordinum*.

Illustrative Fragments

- **Sparta:** Στρατηγία — "the art of generalship," a simplified disciplinary version.
- **Athens:** Πεσσὸς Πολεμικός — "the war-board," a philosophical model of choice.
- **Rome:** *Ludus Ordinum* — "the game teaches formation," integrated into civic discipline.

Key Conclusions

- **Standardization and simplification:** Greco-Roman texts show a trend toward simplification and adaptation for educational and public use.
- **Institutionalization:** mentions of set registration and tournaments reflect formalization and the need for standardized boards.
- **Decline of practical function:** in Roman contexts, the game loses part of its applied

military semantics and becomes partly entertainment, partly ritual.

8.5 Comparison of Texts and Archaeological Data

Summary Table

Text Type	Key Motifs	Archaeological Correspondence
Mesopotamian legends & records	sacredness; military modeling; unit accounting	elite temple pieces; supply records
Egyptian inscriptions	ritual order; royal context	temple boards; inlaid elite sets
Phoenician catalogues	commerce; abstract pieces; maritime theme	mixed port sets; naval iconography
Hellenistic texts	standardization; public tournaments	marble 8×8 boards; match notations
Greek & Latin mentions	education; bureaucratization; decline	late ritual finds; rare elite sets

Synthesis and Critical Notes

- **Complementarity:** texts provide motivation and context; archaeology provides material confirmation. Together they allow reconstruction of functions, rules, and social roles.
- **Indirectness and fragmentation:** many texts are legendary or metaphorical; no complete “rulebook” survives. Reconstruction relies on indirect clues and experimental verification.
- **Regional and chronological variation:** texts confirm what archaeology shows — the game evolved, adapted, and varied locally. No single canonical version existed across all regions and periods.

Conclusion to Section 8

The written tradition of **Kar-Ta-Na** is a multilayered corpus where myth, ritual, military practice, and commerce intertwine.

Mesopotamian and Egyptian legends emphasize the sacred-practical nature of the game; Phoenician and Hellenistic texts document commercial and maritime adaptation; Greco-Roman sources show standardization and gradual functional shift.

The combined analysis of texts and archaeology provides a working foundation for rule reconstruction while leaving room for further discoveries and experimental testing.

9. Final Reconstructed Rules of the Game “Kar-Ta-Na”

Below is a working, experimentally usable summary of the rules of **Kar-Ta-Na**, compiled from archaeological, paleographic, and textual evidence and presented as a playable system. This is a **reconstruction**, not a canon; it is intended for testing and further calibration.

9.1 Structured Full Rule Set

Game Components

- **Board** — a point-grid of nodes and connecting lines; the standard version uses an 8×8 grid of nodes.
 - **Pieces** — double-sided tablets with graphemes arranged top-to-bottom: orientation → state → movement → defense → weapon/range → identifier.
One side shows the **Marching State** (triangle), the other the **Combat State** (rectangle or circle).
 - **Marching Column** — an ordered reserve (stack of pieces); the top piece is visible.
 - **Stochastic element** — a bag with two colors of beads for resolving conflicts.
 - **Resource markers** — logistics tokens (recommended 3–5 per player) for paid deployments and special actions.
-

Game Setup

1. **Board**: place the 8×8 grid; mark edge nodes as deployment points.
 2. **Columns**: each player forms a column of 12–20 slots; the top piece is revealed.
 3. **Initial placement**: either a standard starting layout or free deployment during the first N turns.
Recommended for testing: each player deploys 3 pieces during the first two turns.
-

Turn Structure

Each turn consists of three phases in order:

1. **Movement Phase** — movement along lines, turning, deployment from the column.
 2. **Action Phase** — attacks, special abilities, flipping into combat state.
 3. **Administrative Phase** — replenishment, repairs, logistics, registering effects.
-

Movement Phase — Rules

- **Movement:** a piece moves along edges a number of nodes equal to its **movement** value (M).
Turning at nodes is free unless local rules specify otherwise.
 - **Passing through nodes:** passing through one's own stack is allowed; passing through an enemy stack is forbidden and triggers combat if attempted.
 - **Jumping:** allowed only if the piece bears a special jump symbol; jumping over one occupied node consumes the entire turn.
 - **Deployment from the column:**
 - the top piece is placed on a free edge node in marching state;
 - deployment is part of the movement phase;
 - deployment into the interior may require a logistics token.
-

Action Phase — Rules

- **Flipping:**
 - flipping into combat state usually requires an action;
 - flipping into marching state is typically done during movement;
 - a piece flipped into combat state during the action phase cannot move until the next turn.
 - **Attack:**
 - attacker declares target and attack type;
 - range and orientation are checked;
 - modifiers (flank/rear/fortification) are applied.
 - **Special actions:**
 - general's orders, fortification, repair, trade operations, etc.;
 - most require an action or a resource.
 - **Combat resolution:**
 - attacker's value (a) and defender's value (d) are modified to (a') and (d');
 - the bag receives (a') attacker beads and (d') defender beads;
 - one bead is drawn: attacker color = success; defender color = defense holds;
 - on success, remove one "life" (top plate of the stack) or eliminate the piece.
-

Additional Rules and Clarifications

- **Life stacks:** identical tablets may be stored as a stack; each successful attack removes the top plate.
- **Group attacks:** attack values of participating pieces are summed; resolution follows the same bead-bag procedure.
- **Range:** strokes and arrows indicate radius; firing at maximum range applies a penalty.
- **Victory conditions:**
 - elimination of all enemy units;
 - capture of key nodes;

- reaching a designated point.

Tournament play may use a point system based on kills and node control.

9.2 Rule Variants by Historical Period

Early Bronze Age — Working Reconstruction

- **Board:** variable grids (6×6, 7×7); fortified nodes.
- **Column:** smaller reserve, freer deployment.
- **Randomness:** simple single-draw resolution; special actions rare.
- **Function:** strong ritual and educational component; rules allow local ritual modifications.

Classical Phoenician and Hellenistic Variants

- **Board:** standardized 8×8; marble and inlaid boards.
- **Column:** dynamic, with a market for reinforcements; top piece visible; registration of top N pieces before tournaments.
- **Mechanics:** logistics tokens, catalogues of pieces, naval modifications.
- **Randomness:** multi-draw and success-counting variants for smoother probability curves.

Late Roman Adaptations

- **Board:** 8×8 canonized; boards for public tournaments.
- **Rules:** simplification and bureaucratization; mandatory set registration; standardized notation.
- **Function:** game loses part of its military utility, becomes entertainment and ritual object; elite sets rare due to metal recycling.

9.3 Possible Lost Elements

- **Advanced collecting economy:** detailed rules for buying, trading, and auctioning pieces likely existed as local practices.
- **Campaign system:** rules for long-term campaigns where pieces and columns persist across sessions, including trophies, repairs, and logistics.
- **Complex multiparameter cards:** additional tablets tracking weather, morale, supply, and administrative orders.
- **Formalized ritual procedures:** precise rituals for bead drawing, offerings, or divination linked to game outcomes.
- **Complete match notation:** a fully standardized writing system for recording games; only fragments survive.

- **Local special symbols:** several minor pictograms remain undeciphered (wavy lines, framed dot series).
 - **Multilevel tournaments and regulations:** detailed rules for registration, appeals, and adjudication in public competitions.
-

Conclusion of the Final Reconstruction

The proposed rule summary is a pragmatic, testable model that integrates material evidence, textual hints, and gameplay logic.

It provides a working foundation for creating replicas, running experimental matches, and comparing results with archaeological traces.

The main open questions concern exact modifier values, rare special symbols, and the full mechanics of collecting and ritualization — all promising areas for further field testing and textual analysis.

10. Conclusion

10.1 Significance of the Discovery

Cultural Significance

The reconstruction of **Kar-Ta-Na** reveals the complex interplay between gaming, military practice, and ritual in ancient societies.

The game functioned not merely as entertainment but as a **technology of decision-making**, a medium for transmitting tactical knowledge, and a marker of social status.

Scholarly Significance

Systematizing the material and textual evidence yields a working model that links archaeology, paleography, and game theory.

Even fragmentary finds acquire meaning within the reconstructed mechanics: piece parameters, board structure, and the stochastic mechanism explain observed wear patterns and textual annotations.

Practical Significance

The proposed reconstruction serves as a tool for experimental archaeology, museum replicas, and educational programs.

It enables testing hypotheses about military training, trade networks, and ritual use by reproducing gameplay practices under controlled conditions.

10.2 Prospects for Further Research

Textual Analysis

Targeted work with Mesopotamian, Egyptian, and Hellenistic archives is needed to locate additional references, formulas, and catalogue entries.

Comparing new texts with material finds may clarify rules of collecting and commercial practices.

Archaeological and Materials Research

Priorities include targeted excavations in port and temple contexts, pigment and edge-wear analysis on pieces, and metallographic study of bronze and gold tablets.

Radiometric dating and microscopic wear analysis will refine chronology and usage frequency.

Interdisciplinary Projects

An optimal team includes archaeologists, epigraphers, military historians, ethnographers, and game designers.

Collaborative projects could develop standardized testing protocols, find databases, and open catalogues of replicas.

10.3 Experimental Archaeology and Reconstruction of Game Sets

Replica and Playtesting Project

A series of controlled experiments is recommended:

- producing replicas in clay, wood, bronze, and ivory;
- conducting tournaments using reconstructed rules;
- systematically recording matches and statistically analyzing outcomes.

Repeatability and thorough documentation are essential.

Testing Methodology

- **Phases:** replica preparation → pilot matches → large-scale tests with rule variants.
- **Metrics:** maneuverability, frequency of state flips, average turns to remove a stack, influence of column order on outcomes.
- **Documentation:** video recording, match notation, publication of data in an open repository.

Ethical and Conservation Considerations

Work with originals and valuable materials must follow conservation and legal standards:

- avoid encouraging the melting of artifacts;

- collaborate with museums and legal institutions;
- respect cultural context and the rights of source communities.

Final Remark

The reconstruction of **Kar-Ta-Na** is both a scholarly hypothesis and a practical project. It opens new horizons for understanding ancient societies and provides concrete pathways for testing ideas through experimental archaeology. The next step is organizing an interdisciplinary program of replicas and field tests to transform the working model into an empirically grounded history of the game.

Appendices

Below is a set of auxiliary materials for further work, field testing, and publication: **parameter tables**, **board diagrams**, recommendations for **photographic documentation**, and a set of proposals for **comparative charts**. All materials are formatted for export into spreadsheets and for use in replica production.

Tables of Recorded Piece Parameters

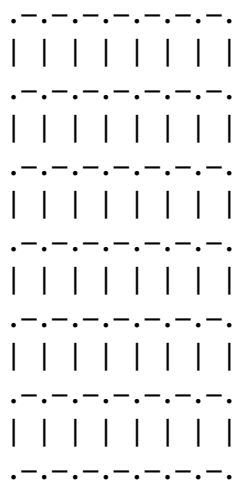
Usage note: each row represents a separate artifact; values are short phrases; avoid line breaks within cells when exporting to CSV/Excel.

ID	Material	Orientation/ Marker	Parameters (M/ A/D/Dist)	Find Context
F-001	clay	arrow north; triangle (march)	M=3; A=1; D=1; Dist=—	domestic layer, Nippur
F-012	ivory	circle (combat); general sign	M=1; A=4; D=3; Dist=1	temple burial
F-023	bronze	arrow east; two strokes (range)	M=2; A=2; D=2; Dist=3	coastal city, trophy
F-037	stone (marble)	triangle; inlay	M=3; A=1; D=2; Dist=—	public mosaic board
F-058	bronze (stamped)	rectangle (combat); wavy line	M=1; A=3; D=2; Dist=2	palace inventory

Reconstructed Board Diagrams

8×8 Standard Grid (Reconstruction)

A minimal visualization of the node grid (each . is a node; |/- are connections).
For printing, use a vector grid with 8×8 nodes and marked edge-deployment nodes.



- Legend:**
- **Edge nodes** — recommended deployment points; mark with a contrasting ring.
 - **Special nodes** — mark with inlay/color for “fortifications” or “objective points.”

Variants by Period (schematic)

- **Early (6×6 / 7×7)** — lower node density; add 1–2 “special” central nodes.
- **Naval variant** — 8×8 with highlighted “corridors” (two parallel lines without cross-links).
- **Roman public slab** — 8×8 with holes for fixing large disks and Latinized node labels.

Artifact Photography (Guidelines and Catalogue Template)

Photography Recommendations

- **File formats:** TIFF for archive; JPEG 3000×2000 px for publication.
- **Naming:** SiteID_ObjectID_Material_View_Date
(e.g., NIP_F001_ivory_top_20260516).
- **Required metadata:** ID, material, dimensions (mm), weight (g), context (layer/site), date, photographer initials.
- **Angles:** overview, close-up of face, edge, reverse, scale bar.
- **Lighting:** diffuse, no glare; side lighting for inlays and relief.

Catalogue Template (one row per artifact)

File	ID	View	Material	Context
------	----	------	----------	---------

File	ID	View	Material	Context
NIP_F001_top.tif	F-001	face	clay	domestic layer, Nippur
NIP_F001_edge.tif	F-001	edge	clay	domestic layer, Nippur
MAR_F023_overview.jpg	F-023	overview	bronze	coastal city

Note: do not publish original photographs without museum/owner permission; store rights and sources in the repository.

Comparative Diagrams (Proposals for Visualization)

Purpose of the diagrams:

to illustrate correlations between material, context, and piece parameters; to visualize the evolution of the board; and to track the frequency of special symbols.

Recommended Charts and Input Tables

1. Histogram of Material Distribution

- **Input table:** Material; Count
- **Purpose:** to show the proportion of clay/stone/bronze/ivory pieces.

2. “Material vs. Average Attack” Diagram

- **Input table:** Material; Mean_A
- **Purpose:** to test the hypothesis that elite materials correlate with higher attack values.

3. Heatmap of Node Usage by Epoch

- **Input table:** Epoch; NodeX; NodeY; Frequency
- **Purpose:** to visualize wear concentration and commonly used positions.

4. Network Graph of “Exchanges and Trophies”

- **Input table:** SourceSite; DestinationSite; Count
- **Purpose:** to show trade and trophy flows of pieces between regions.

Export format:

CSV for tables; SVG/PNG for diagrams.

Store raw data and generation code (R/Python) in the repository.

Practical Materials for Replicas and Tests

Template Parameters for Replicas

Recommended baseline values:

- **Light infantry:**
M = 1; A = 1; D = 1; Dist = —
- **Heavy infantry:**
M = 1; A = 2; D = 3; Dist = —
- **Cavalry:**
M = 4; A = 2; D = 1; Dist = —
- **Archers:**
M = 1; A = 1; D = 1; A = 2; Dist = 2,4 (penalty at max range)

Recommended Materials for Replica Production

- **Clay** — low cost, suitable for large batches.
- **Wood** — medium cost, durable, easy to carve.
- **Bronze/Brass** — elite replicas, historically accurate for high-status sets.
- **Ivory** — replace with an ethical modern analogue (e.g., layered bone-composite).

Appendix Conclusion

The appendices provide a practical toolkit for documenting finds, producing replicas, and conducting statistical and visual analyses.

If you want, I can prepare:

- ready-to-use **CSV templates** for all tables,
- a clean **vector 8×8 grid** in SVG format,
- and a **sample Python/Matplotlib script** for generating histograms and heatmaps.